

Network Theory

BAECE103

Question Bank

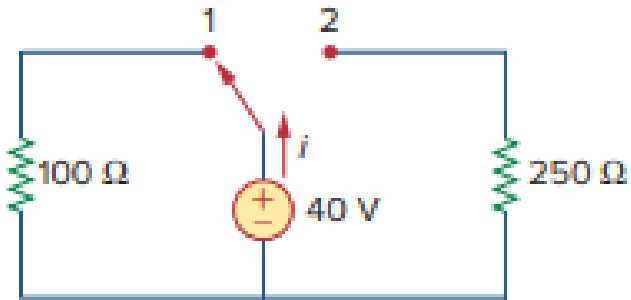
Fall 2-25-26

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Professor, School of Electronics Engineering

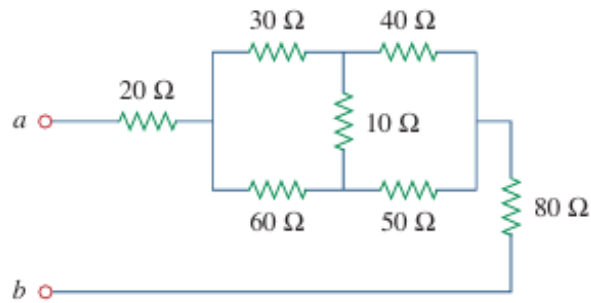
Module:1

DC Circuit Analysis and Theorems: Basic Circuit Elements and Sources; Ohm's Law; Kirchhoff's Laws; Series Resistors and Voltage Division; Parallel Resistors and Current Division; Wye-Delta Transformations; Methods of Analysis Nodal Analysis, Mesh Analysis; Theorems- Superposition Theorem, Source Transformation Theorem, Thevenin's Theorem, Norton's Theorem, Maximum Power Transfer Theorem, Reciprocity Theorem.

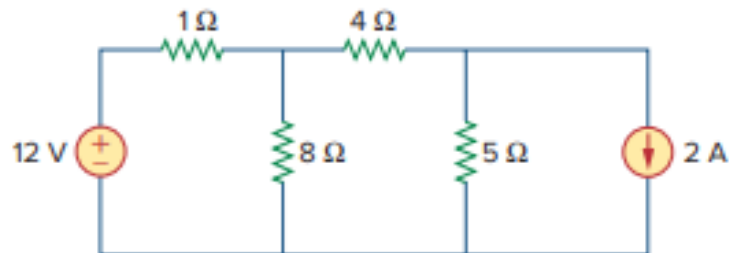
1. Find the current I when the switch is in position 1 and then when in position 2.



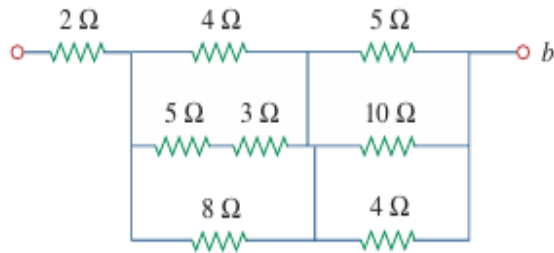
2. Obtain the equivalent resistance R_{ab} between terminals a-b in the circuit shown



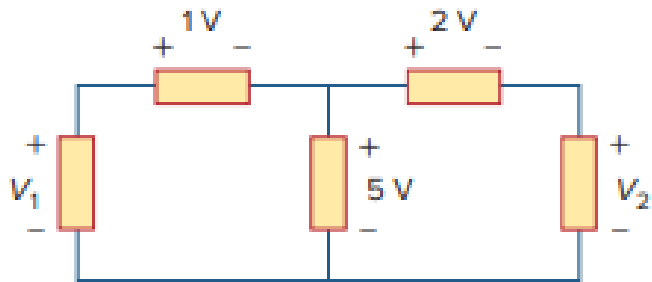
3. Determine the number of branches and nodes in the circuit given below.



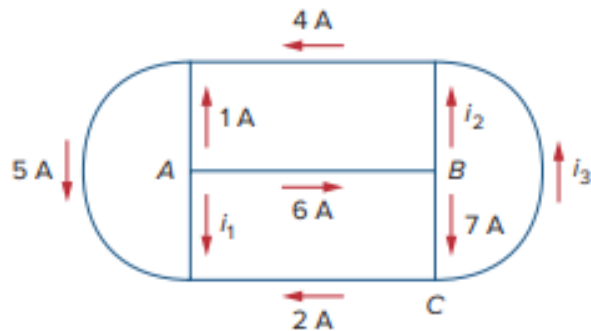
4. Reduce the circuit shown to a single resistor at terminals a-b.



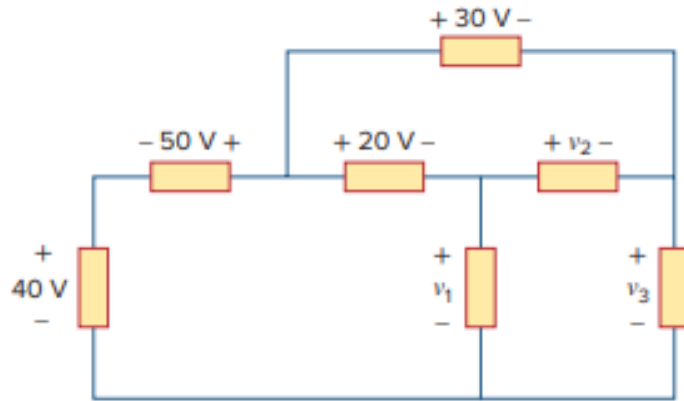
5. Calculate V_1 and V_2 .



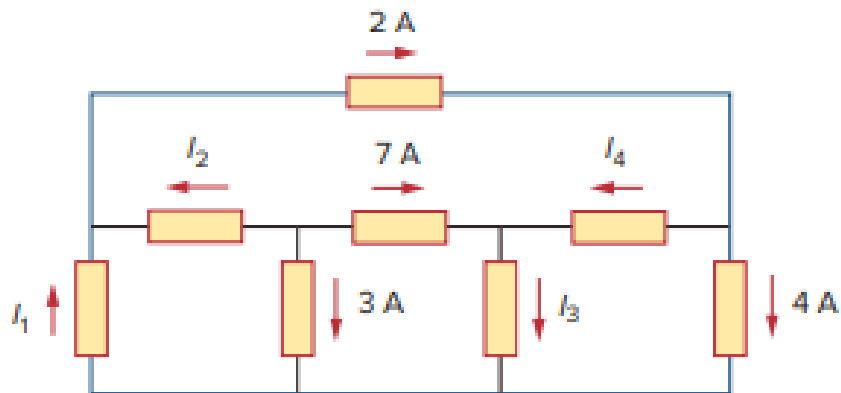
6. Find the currents i_1 , i_2 , i_3 .



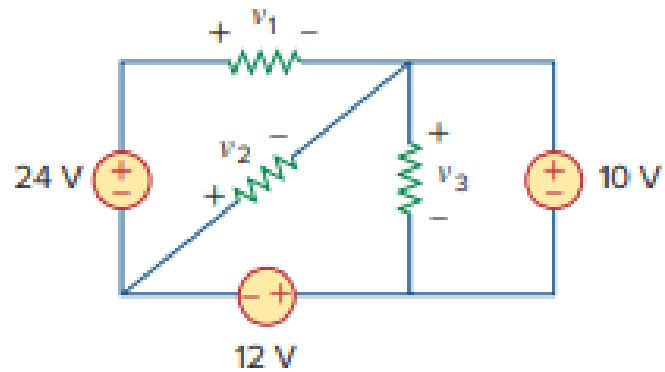
7. Calculate V_1 , V_2 and V_3



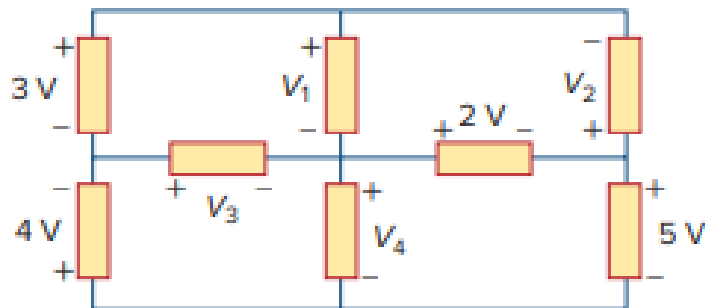
8. Find currents i_1 to i_4 .



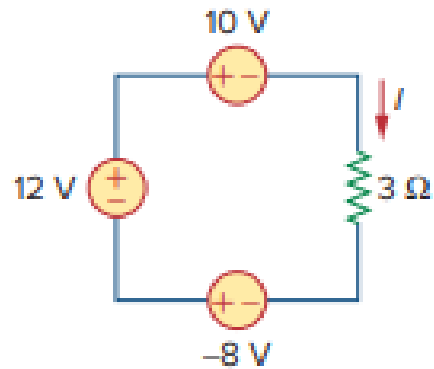
9. Determine V_1 through V_3 in the circuit.



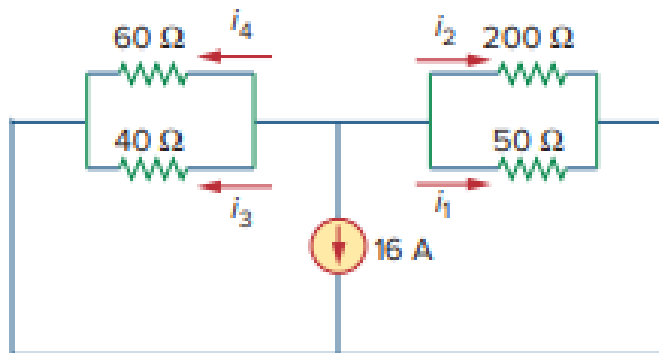
10. Calculate V_1 to V_4 using KVL.



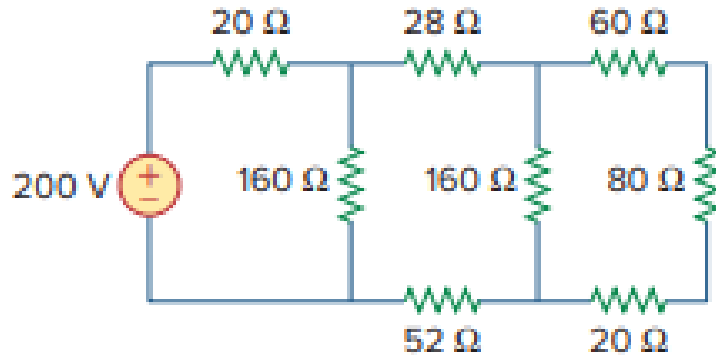
11. Find I , the power dissipated by the resistor, and the power supplied by each source.



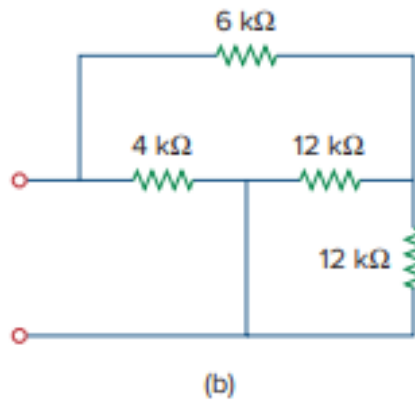
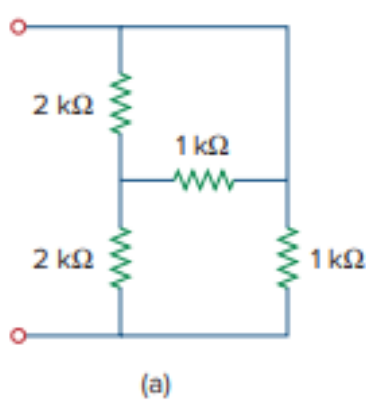
12. Calculate i_1 to i_4 in the circuit.



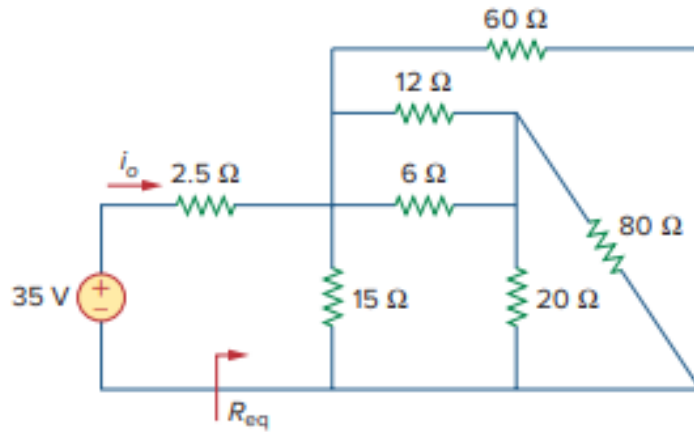
13. Using series/parallel resistance combination, find the equivalent resistance seen by the source in the circuit. Find the overall absorbed power by the resistor network.



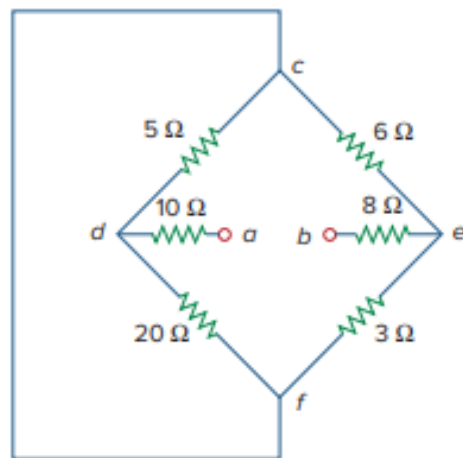
14. Evaluate R_{eq} looking into each set of terminals for each of the circuits.



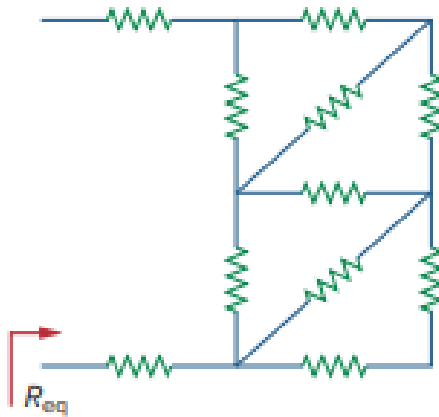
15. Find R_{eq} and i_o in the circuit.



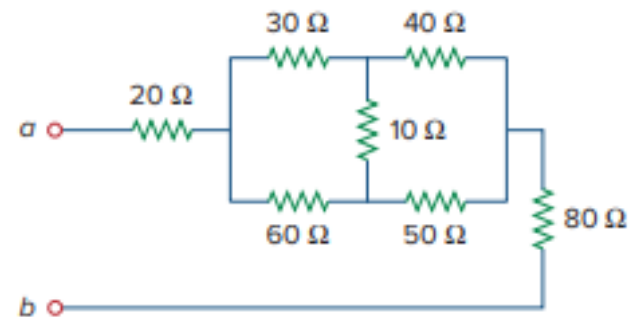
16. Find the equivalent resistance R_{ab} in the circuit.



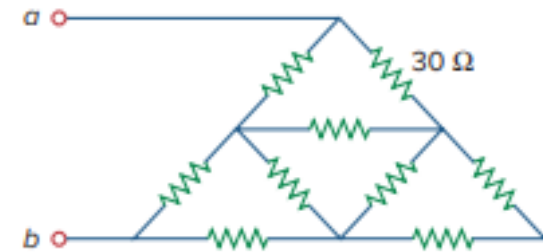
17. For the circuit shown below, find the equivalent resistance. All resistors are 3Ω .



18. Obtain the equivalent resistance R_{ab} in each of the circuits (a) and (b). In (b), all resistors have a value of 30Ω .

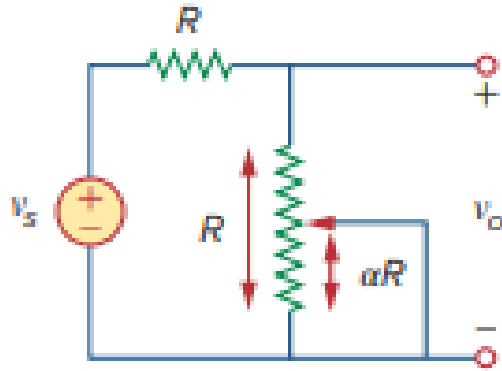


(a)

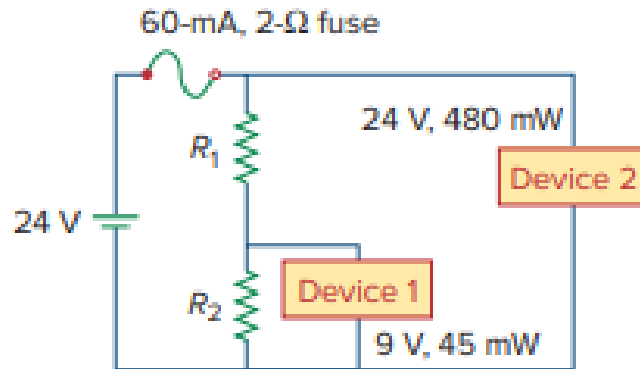


(b)

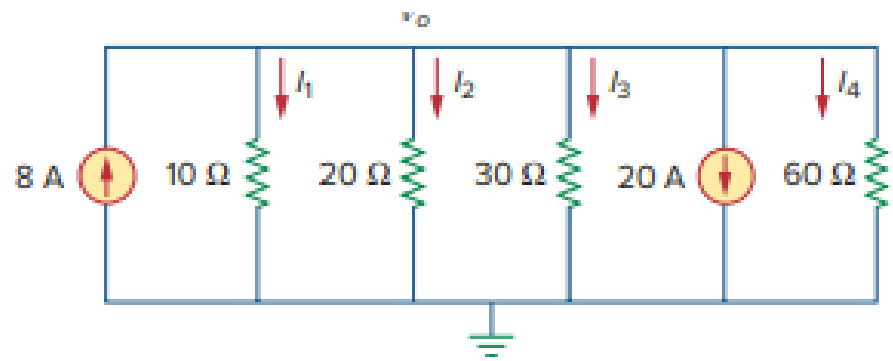
19. In the circuit, the wiper divides the potentiometer resistance between αR and $(1 - \alpha)R$, $0 \leq \alpha \leq 1$. Find v_o / v_s .



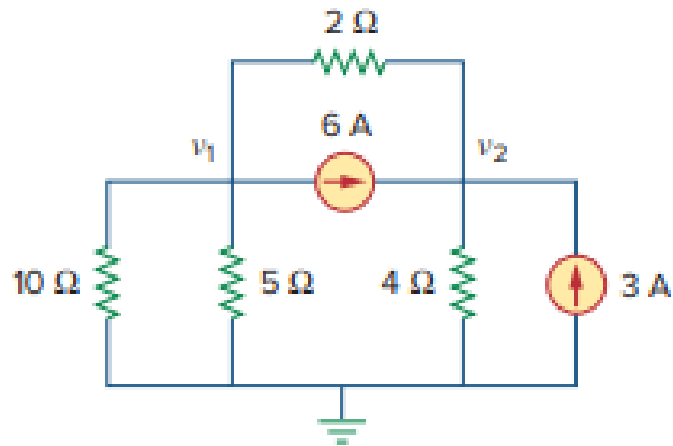
20. Two delicate devices are rated as shown. Find the values of the resistors R_1 and R_2 needed to power the devices using a 24-V battery.



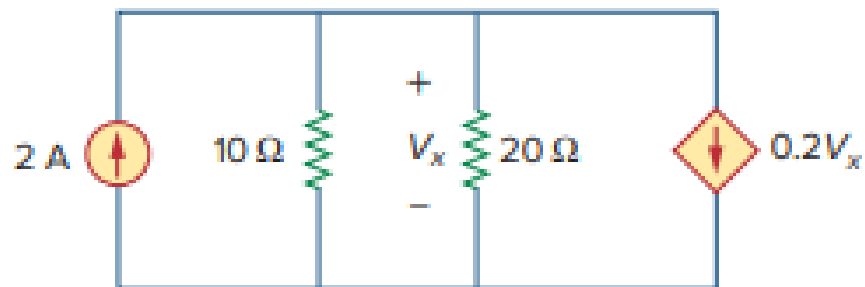
21 . Find the currents I_1 through I_4 and the voltage v_o in the circuit using Nodal analysis method.



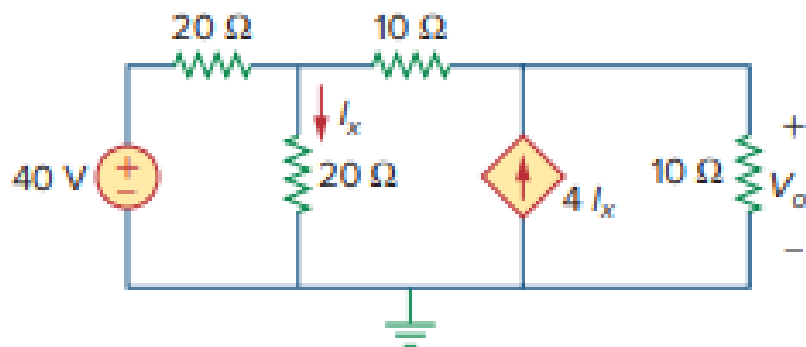
22. Find the voltages v_1 and v_2 using the nodal analysis method.



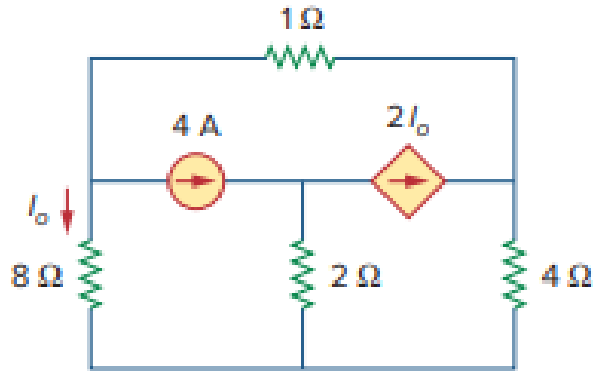
23. Apply nodal analysis to solve for V_x in the circuit.



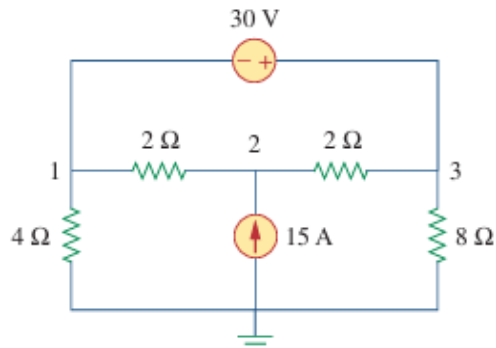
24. Using nodal analysis, determine V_o in the circuit.



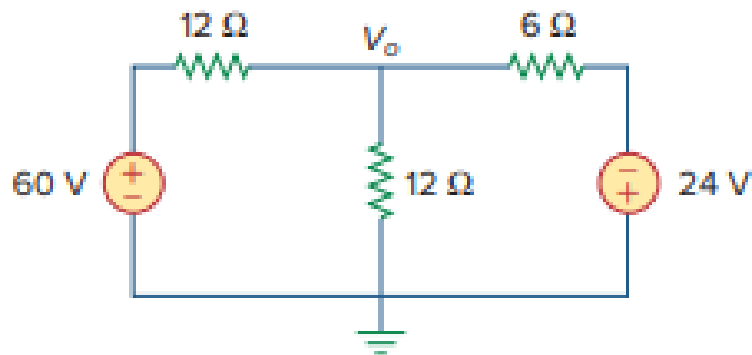
25. Find I_o in the circuit using Nodal analysis method.



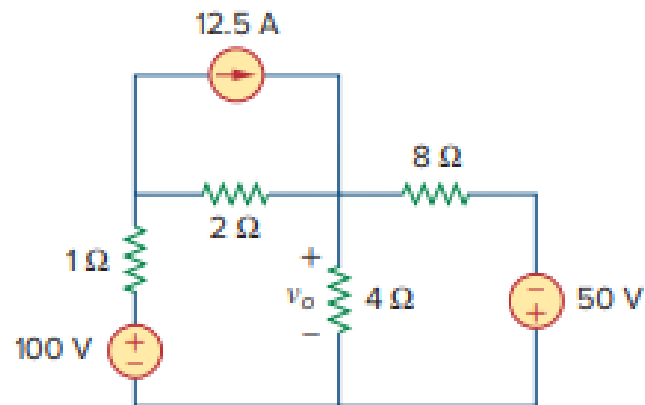
26. Determine the node voltages in the circuit given the circuit using nodal analysis method.



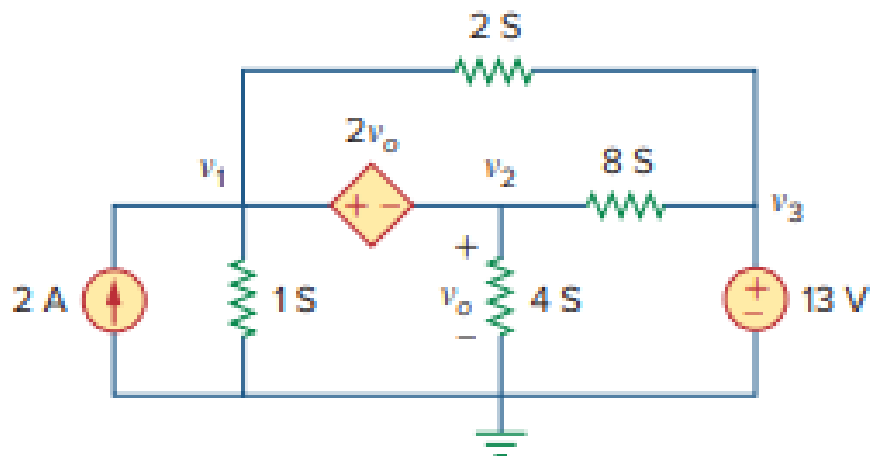
27. Find V_o and the power dissipated in all the resistors in the circuit. Use Nodal analysis method.



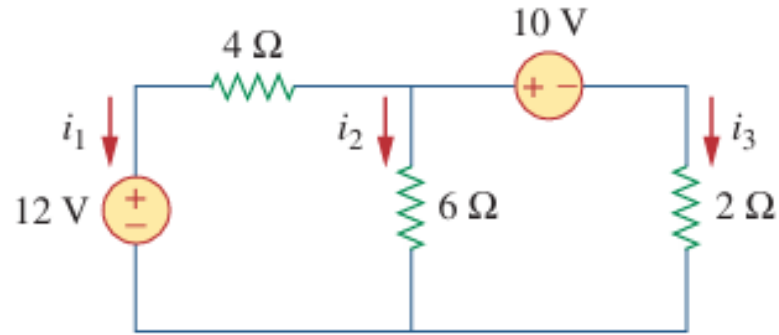
28. Using nodal analysis, find v_o in the circuit.



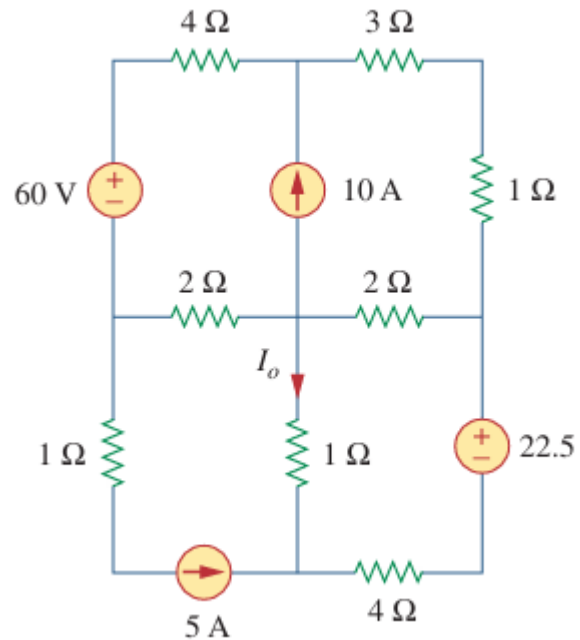
29. Determine voltages v_1 through v_3 in the circuit using nodal analysis.



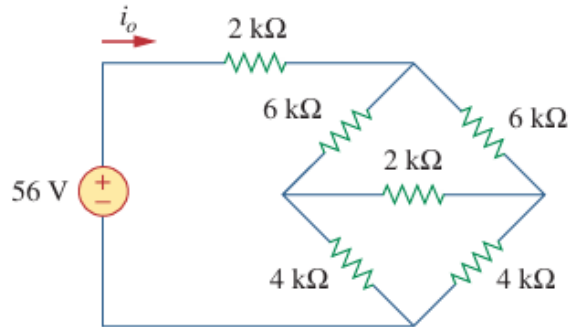
30. Use mesh analysis to obtain i_1 , i_2 , and i_3 in the circuit



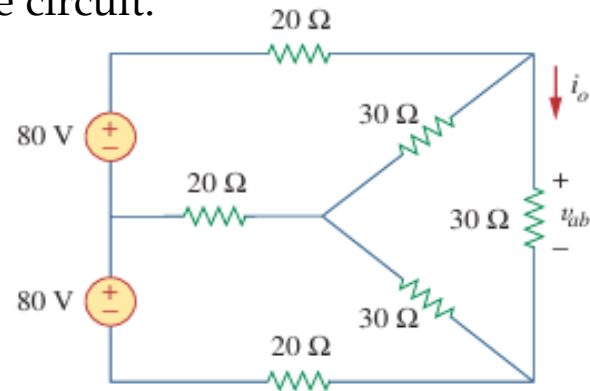
31. Apply mesh analysis to the circuit and obtain I_o .



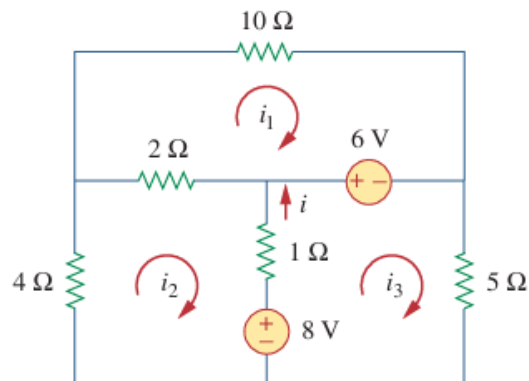
32. For the bridge network, find i_o using mesh analysis.



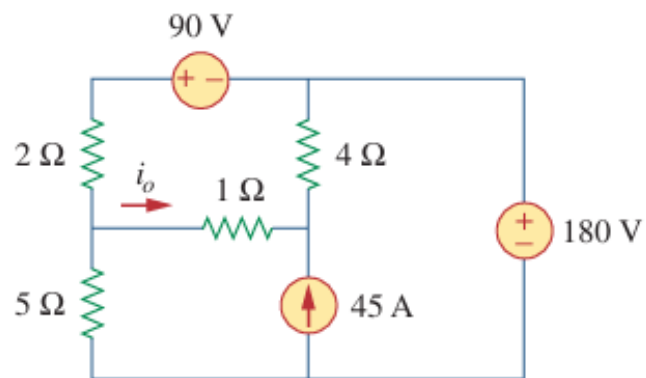
33. Use mesh analysis to find v_{ab} and i_o in the circuit.



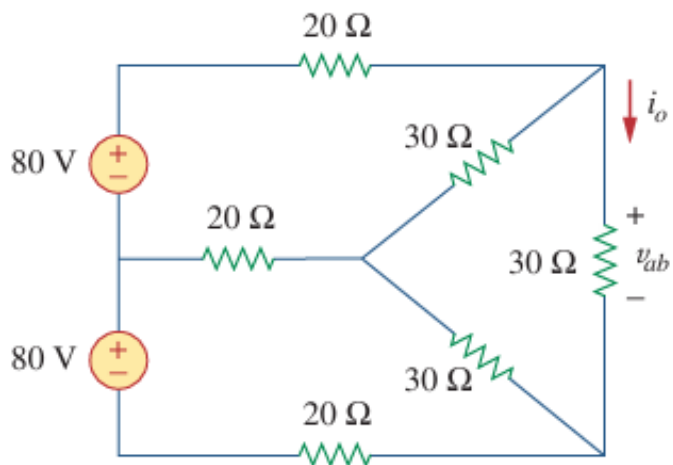
34. Apply mesh analysis to find i .



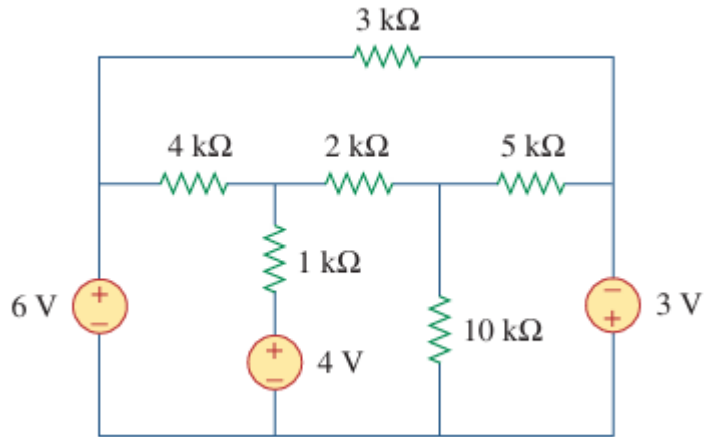
35. Use mesh analysis to obtain in the circuit.



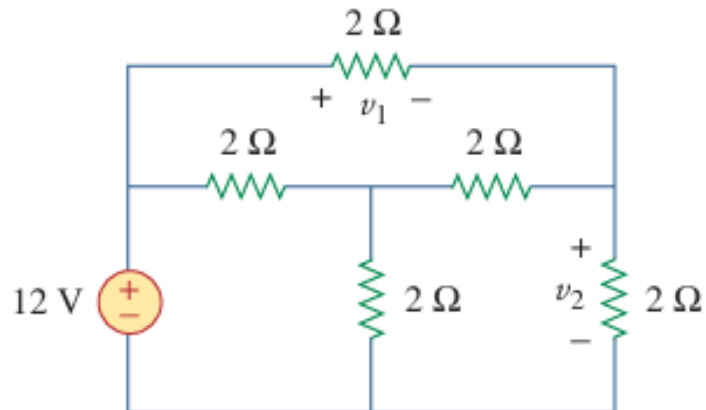
36. Use mesh analysis to find v_{ab} .



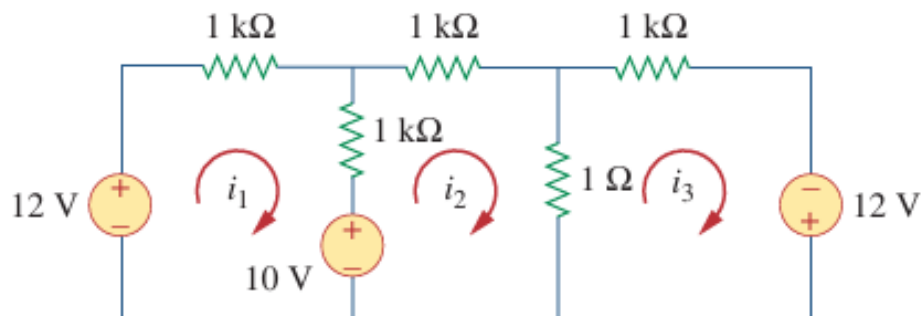
37. Determine the current through the 10-k resistor in the circuit using mesh analysis.



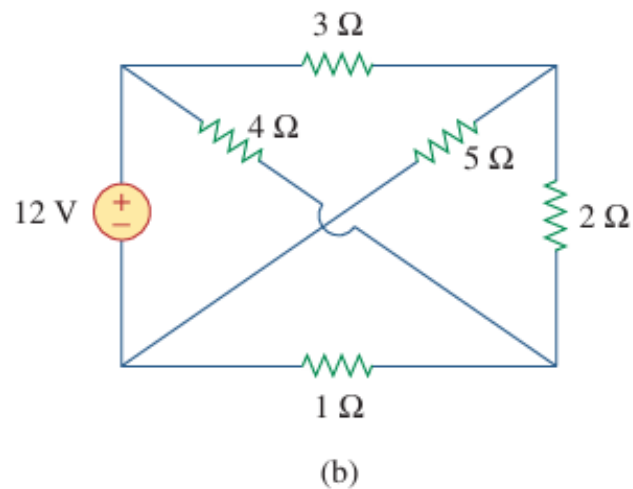
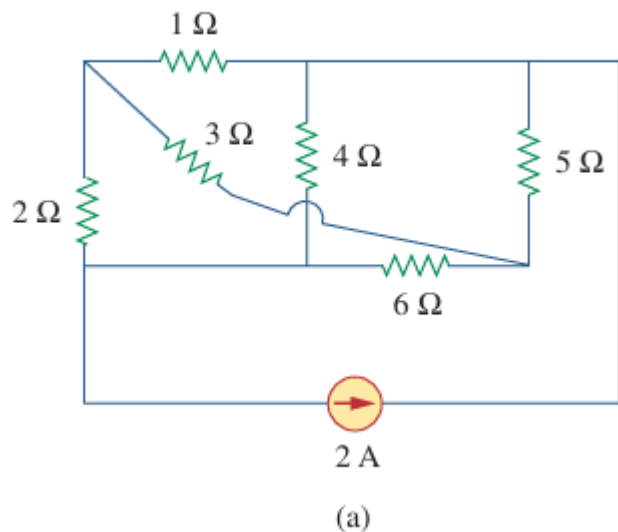
38. Determine voltages v_1 and v_2 in the circuit using mesh analysis.



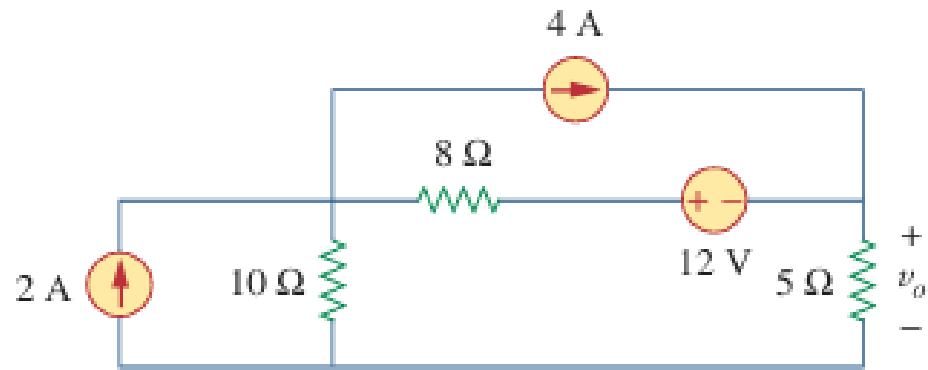
39. Find the mesh currents i_1 , i_2 , and i_3 in the circuit.



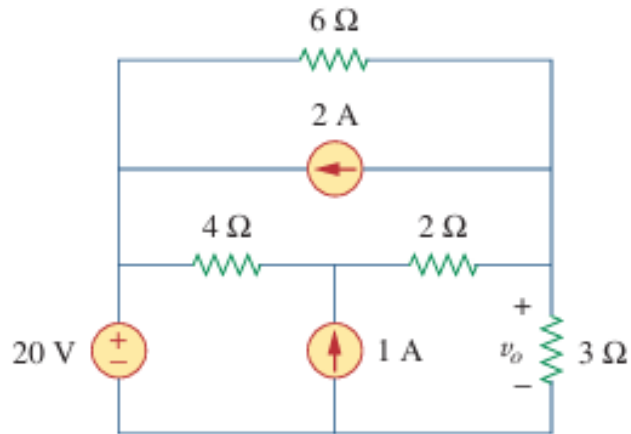
40. Which of the circuits is planar? For the planar circuit, redraw the circuits with no crossing branches.



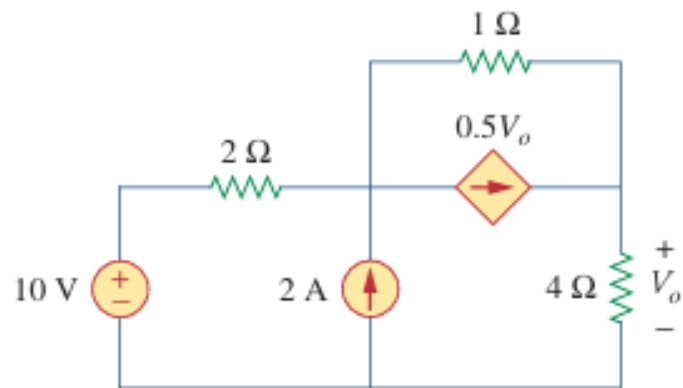
41. Using Superposition theorem find v_o .



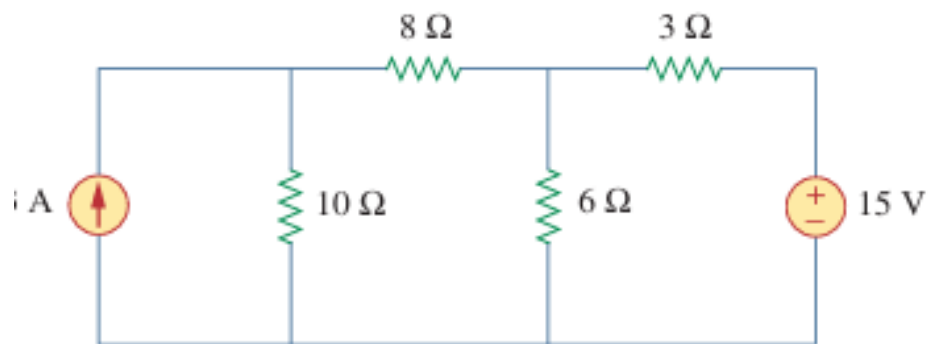
42. Apply the superposition principle to find v_o circuit .



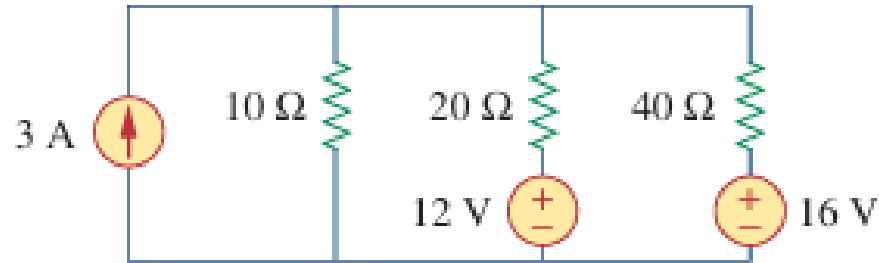
43. Use Superposition to find in the circuit.



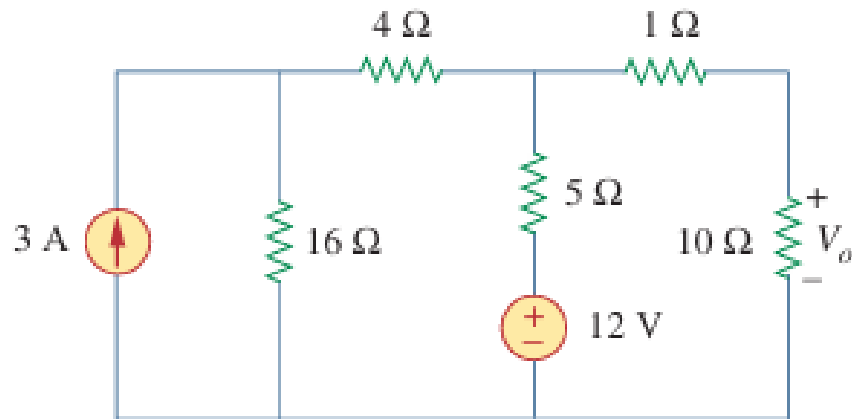
44. Referring to the circuit below, use source transformation to determine the current and power absorbed by the $8\ \Omega$ resistor..



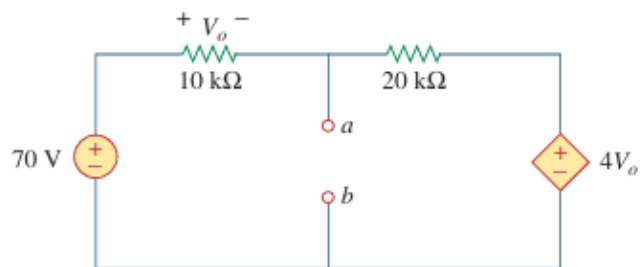
45. Use source transformation to reduce the circuit below into a single voltage source in series with a single resistor.



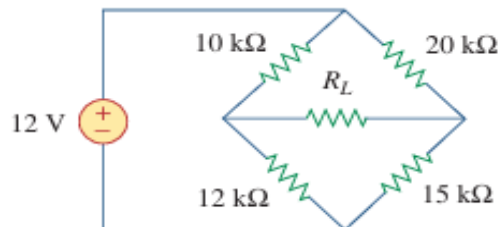
46. Apply Thevenin's theorem to find V_o in the circuit.



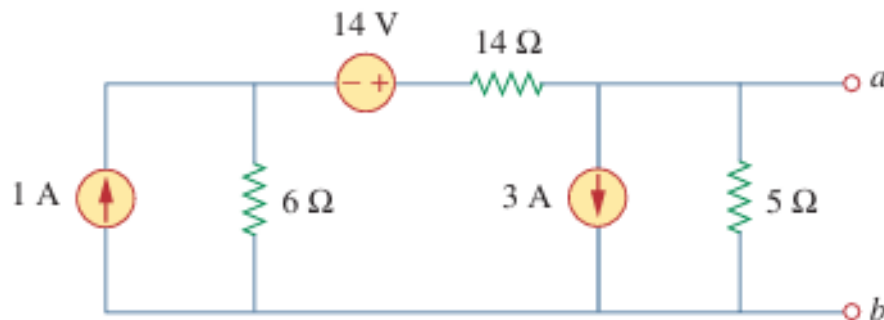
47. Find the Thevenin's equivalent at terminals a-b of the circuit.



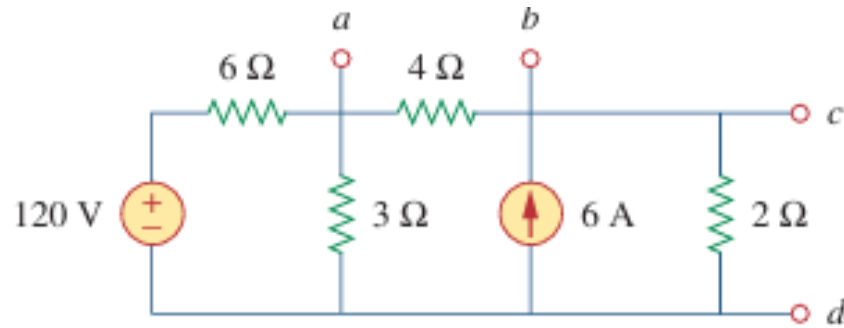
48. Consider the circuit given. (a) Replace the resistor by a zero resistance ammeter and determine the ammeter reading. (b) To verify the reciprocity theorem, interchange the ammeter and the 12-V source and determine the ammeter reading again.



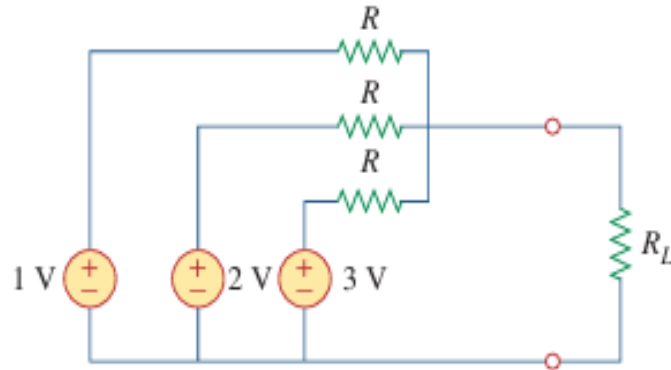
49. Find the Thevenin and Norton equivalents at terminals of the circuit.



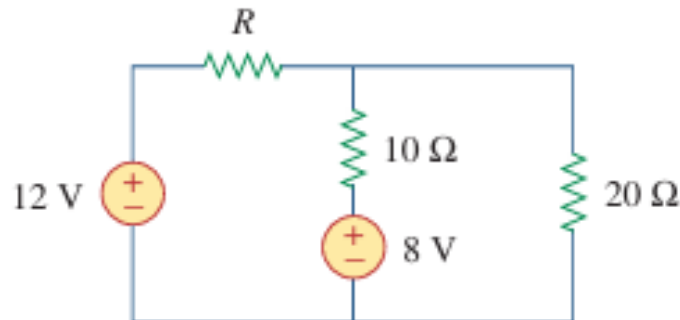
50. Given the circuit, obtain the Norton equivalent as viewed from terminals i) a-b ii) c-d.



51. For the circuit, determine the value of R such that the maximum power delivered to the load is 3 mW.



52. Compute the value of R that results in maximum power transfer to the 10- resistor in circuit. Find the maximum power.

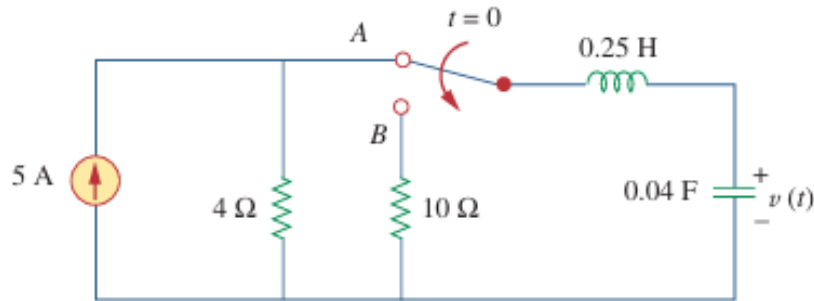


Module:2

Transient Analysis of First and Second-Order Circuits: First-Order Circuits- Source-Free RC Circuit, Source-Free RL Circuit, Singularity Functions, Step Response of an RC Circuit, Step Response of an RL Circuits; Second-Order Circuits- Source-Free Series RLC Circuit, Source-Free Parallel RLC Circuit, Step Response of a Series RLC Circuit, Step Response of a Parallel RLC Circuit, General Second-Order Circuits.

53. If $R = 50\ \Omega$, $L = 1.5\ \text{H}$, what value of C will make an RLC series circuit: (a) overdamped, (b) critically damped, (c) underdamped?

54. In the circuit, the switch instantaneously moves from position A to B at $t = 0$. Find $v(t)$ for all $t \geq 0$.



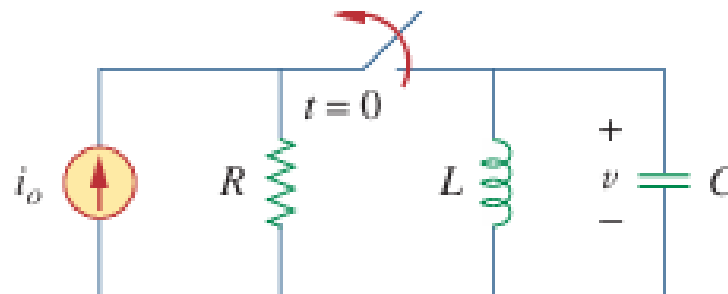
55. The responses of a series RLC circuit are

$$v_C(t) = 30 - 10e^{-20t} + 30e^{-10t}\ \text{V}$$

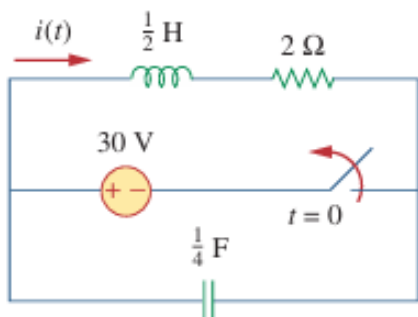
$$i_L(t) = 40e^{-20t} - 60e^{-10t}\ \text{mA}$$

where v_C and i_L are the capacitor voltage and inductor current, respectively. Determine the values of R , L , and C .

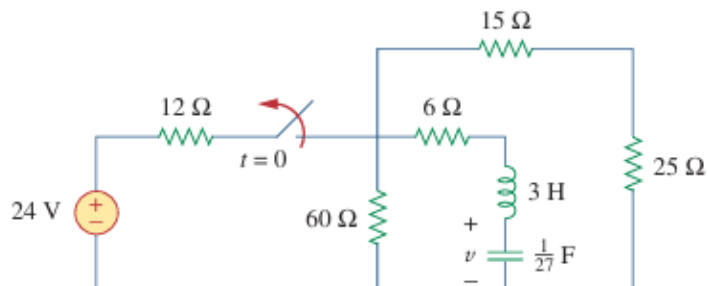
56. Find $v(t)$ for $t > 0$ in the circuit.



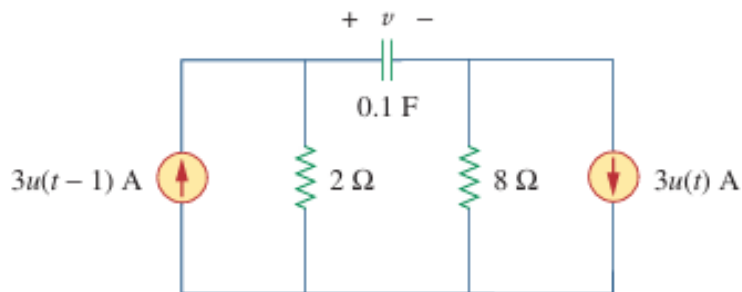
57. The switch in the circuit has been closed for a long time but is opened at $t = 0$. Determine for $i(t)$ for $t > 0$.



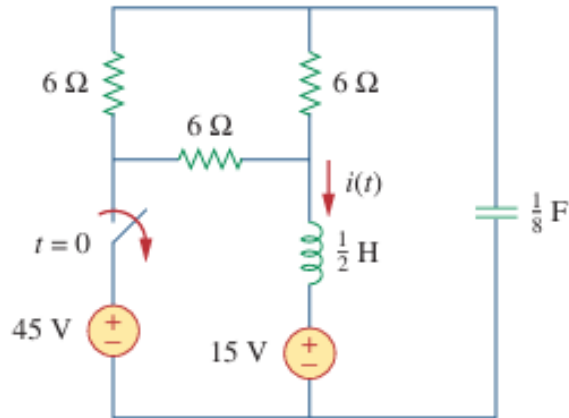
58. Calculate $v(t)$ for $t > 0$ in the circuit .



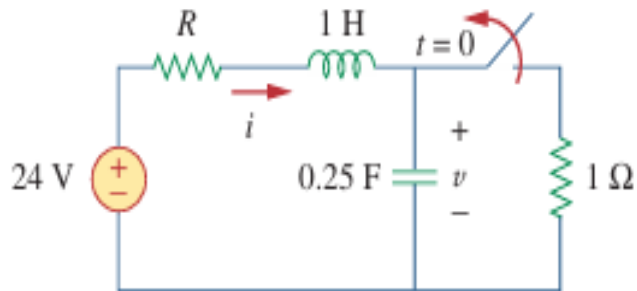
59. Determine $v(t)$ for $t > 0$ in the circuit if $v(0) = 0$.



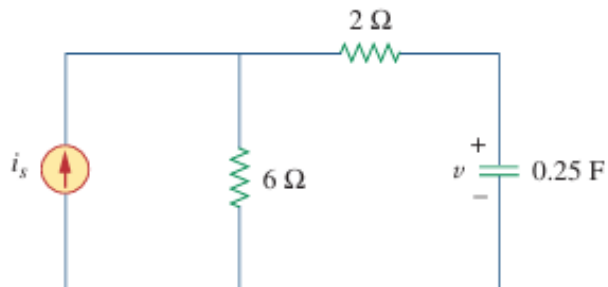
6. For the network shown below, solve for $i(t)$ for $t > 0$.



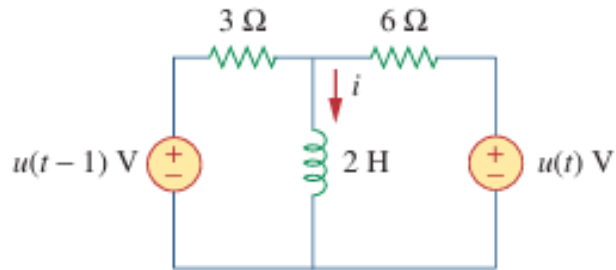
* For the circuit, find $v(t)$ and $i(t)$ for $t > 0$ when $R = 5\text{k}\Omega$.



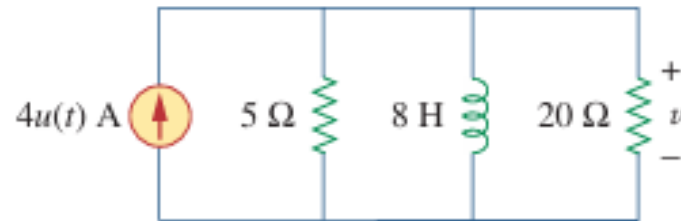
& For the circuit shown below $i_s(t) = 5u(t)$, find $v(t)$.



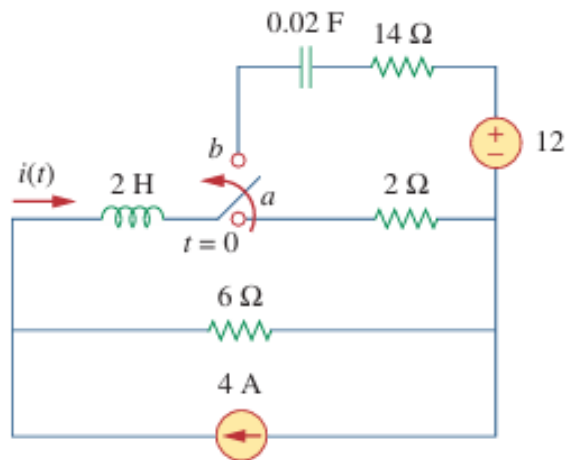
60. For the circuit, calculate $i(t)$ if $i(0)=0$.



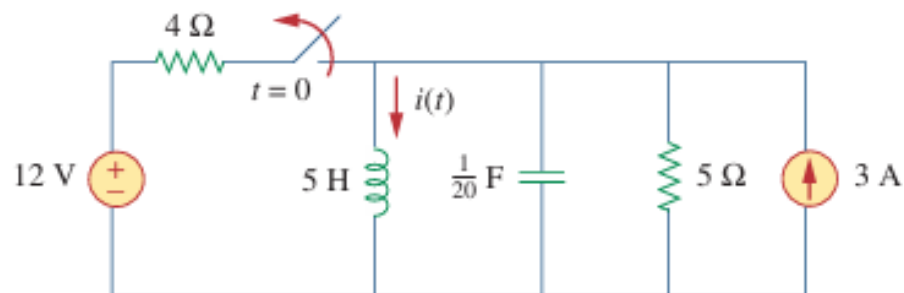
61. Find $v(t)$ for $t > 0$ in the circuit if the initial current in the inductor is zero.



62. The switch in the circuit is moved from position a to b at $t = 0$. Determine $i(t)$ for $t > 0$.

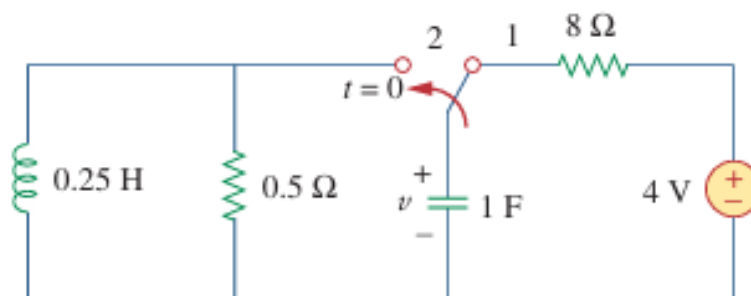


63. Determine $i(t)$ for $t > 0$ in the circuit.

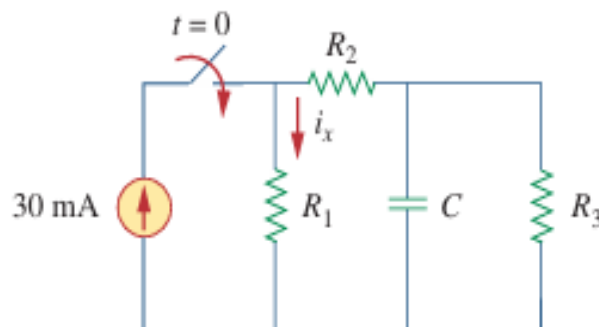


64. In the circuit, the switch has been in position 1 for a long time but moved to position 2 at $t = 0$. Find:

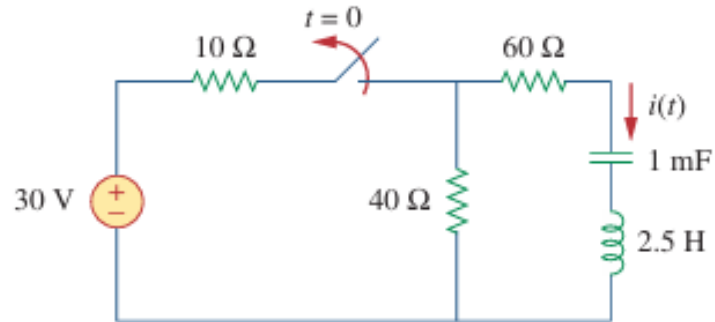
- (a) $v(0^+)$, $dv(0^+)/dt$,
- (b) $v(t)$ for $t \geq 0$.



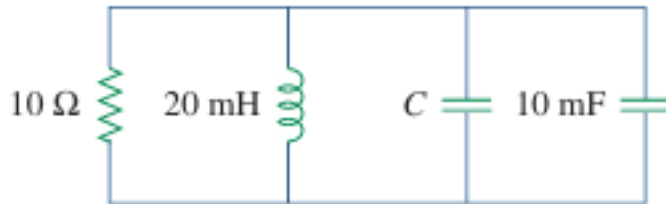
65. In the circuit, find i_x for $t > 0$. Let $R_1 = R_2 = 1 \text{ k}\Omega$, $R_3 = 2 \text{ k}\Omega$ and $C = 0.25 \text{ mF}$ due to the step input applied.



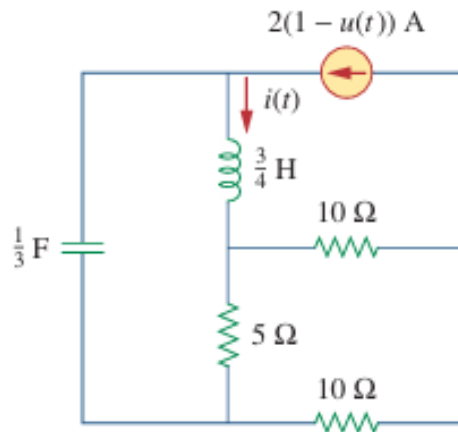
66. Find $i(t)$ for $t > 0$ in the circuit.



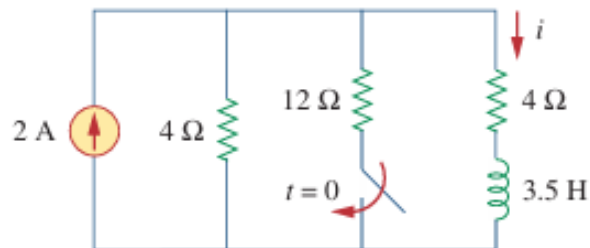
67. For the network, what value of C is needed to make the response underdamped with unity damping factor ($\alpha = 1$)?



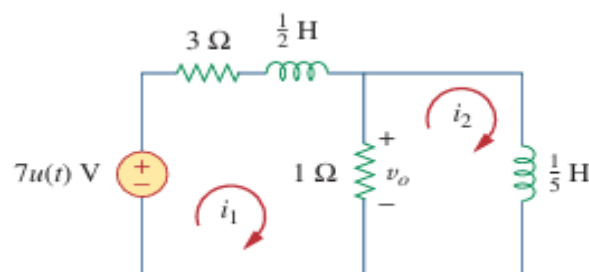
68. Calculate $i(t)$ for $t > 0$.



69. Obtain the inductor current for both $t < 0$ and $t > 0$ in the circuit.

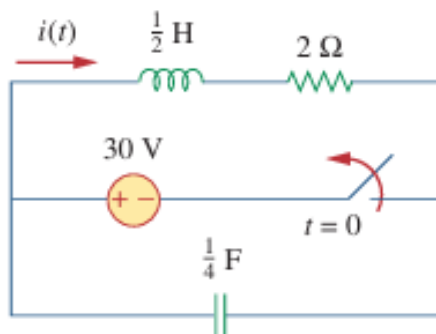


70. Find $v_o(t)$ for $t > 0$ in the second order circuit given

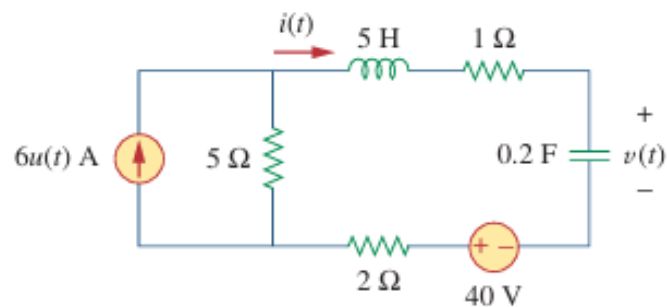


71. A series RLC circuit has $R = 20 \text{ k}\Omega$, $L = 0.2 \text{ mH}$ and $C = 5 \mu\text{F}$. What type of damping is exhibited by the circuit?

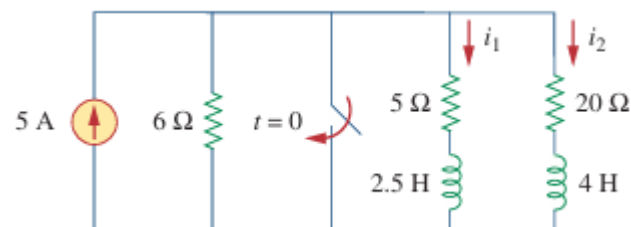
72. The switch in the circuit has been closed for a long time but is opened at $t = 0$. Determine $i(t)$ for $t > 0$.



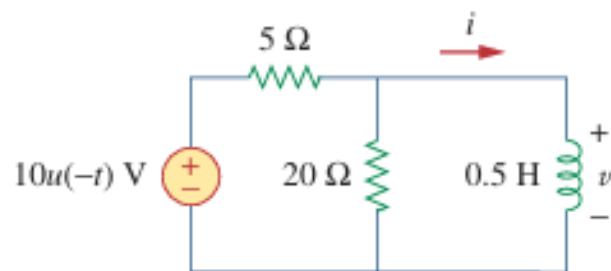
73. Obtain and for in the circuit.



74. Find $i_1(t)$ and $i_2(t)$ for $t > 0$ in the circuit on application of a step input.



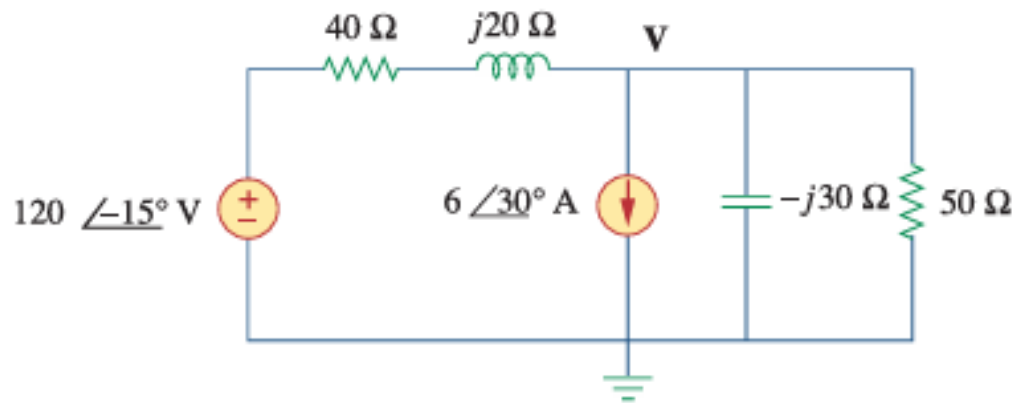
75. Obtain and in the circuit.



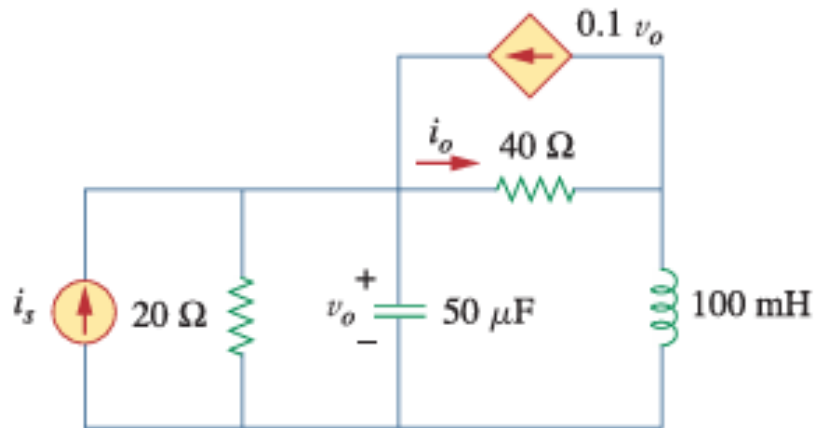
Module:3

Sinusoidal Steady-State Analysis: Sinusoids; Phasors; Phasor Relationships for Circuit Elements; Impedance and Admittance; Kirchhoff's Laws in the Frequency Domain; Instantaneous and Average Power; Complex Power; Methods of Analysis-Nodal Analysis, Mesh Analysis; Theorems-Superposition Theorem, Source Transformation Theorem, Thevenin's Theorem, Norton's Theorem, Maximum Average Power Transfer Theorem.

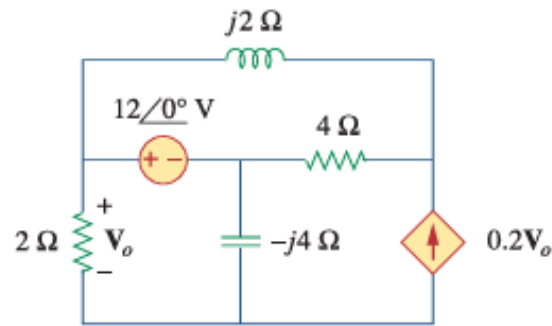
76. Find $\mathbf{V}$ in the circuit using Nodal analysis method.



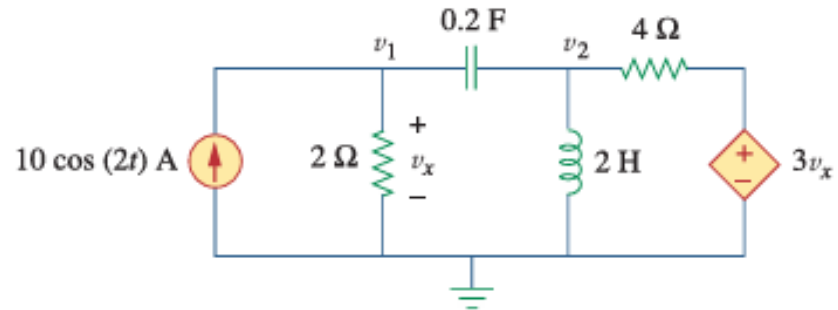
77. Find i_o in the circuit using Nodal analysis method. Let $i_s = 6 \cos(200t + 15^\circ)$



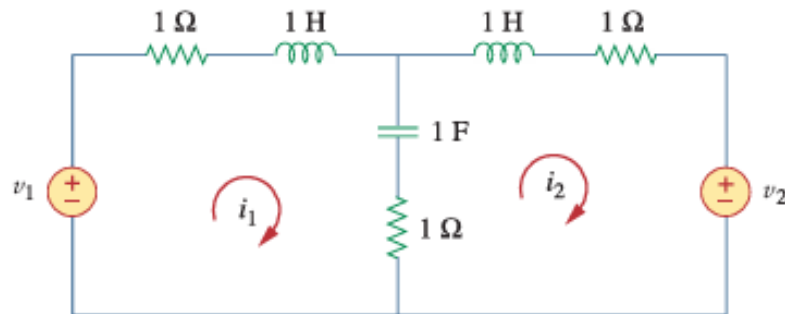
78. Find V_0 in the circuit using Nodal analysis



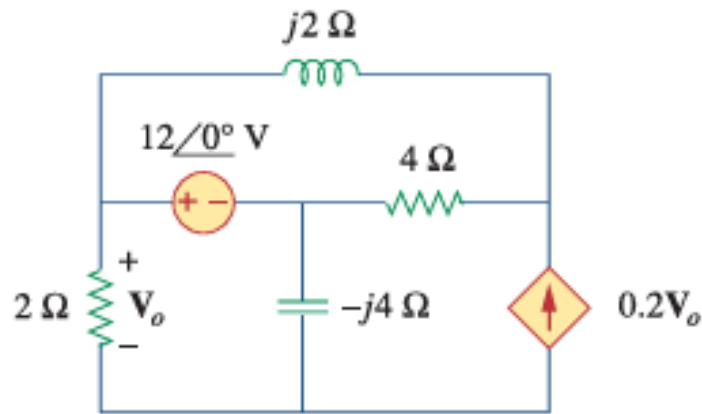
79. Using Nodal analysis find v_1 and v_2 in the circuit.



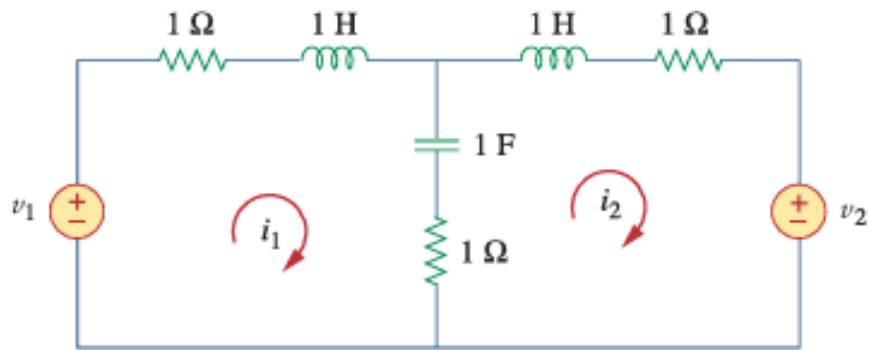
80. In the circuit, determine the mesh currents i_1 and i_2 . Let $v_1 = 10\cos 4t$ V and $v_2 = 20\cos(4t - 30^\circ)$ V.



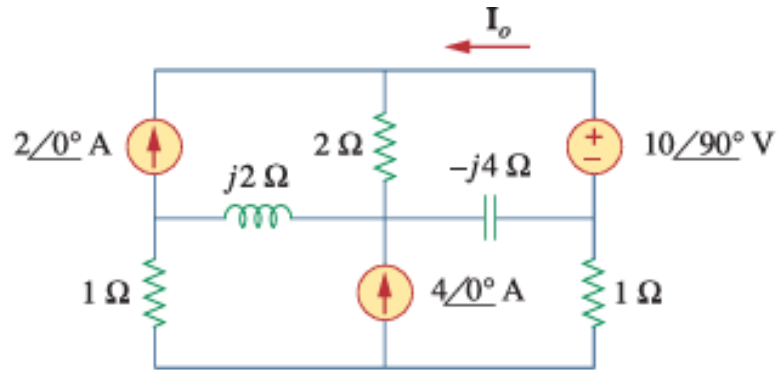
81. Find V_o in the circuit using Nodal analysis method.



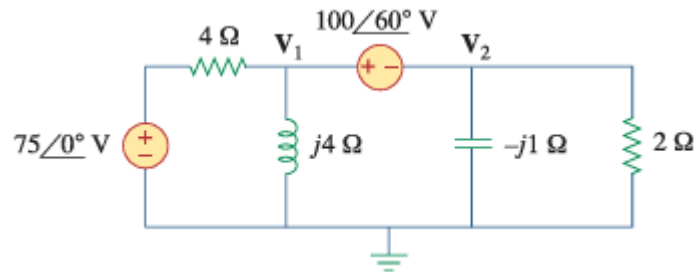
82. In the circuit, determine the mesh currents i_1 and i_2 . Let $v_1 = 10\cos 4t\text{ V}$ and $v_2 = 20\cos(4t - 30^\circ)\text{ V}$.



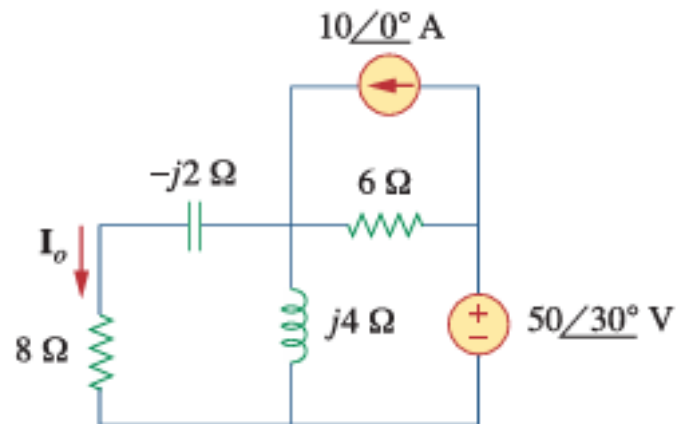
83. Find I_0 in the circuit using Mesh analysis method.



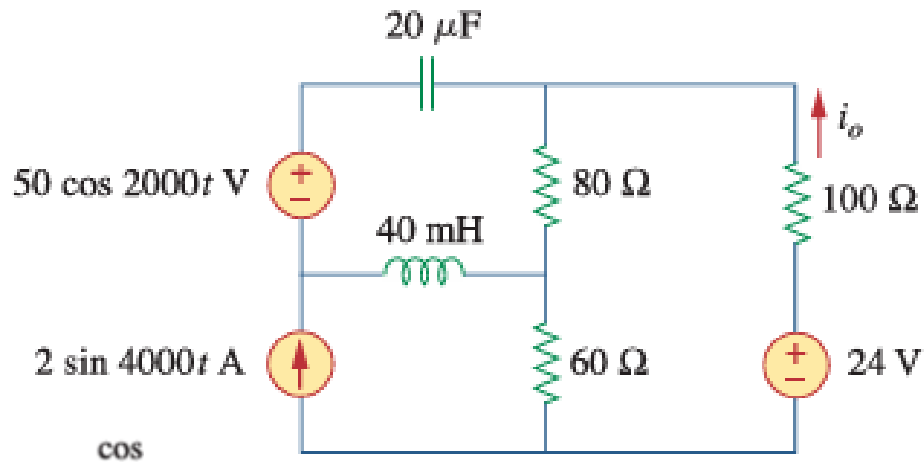
84. Calculate V_1 and V_2 in the circuit.



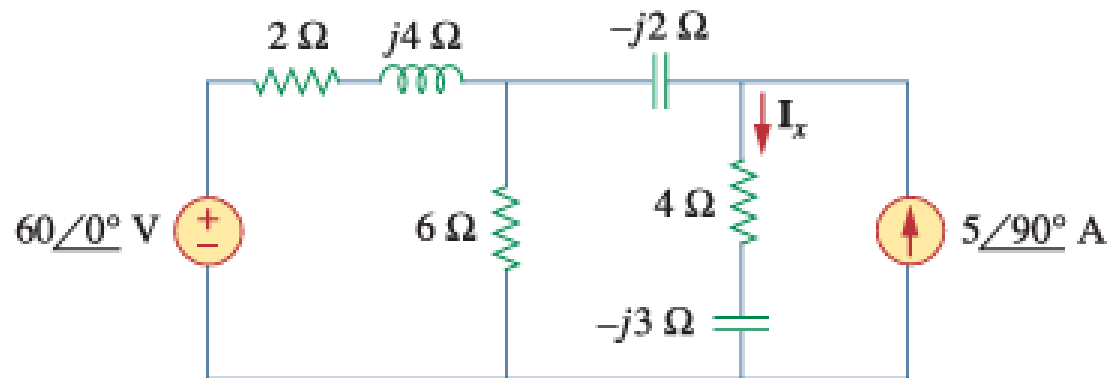
85. Find I_0 in the circuit.



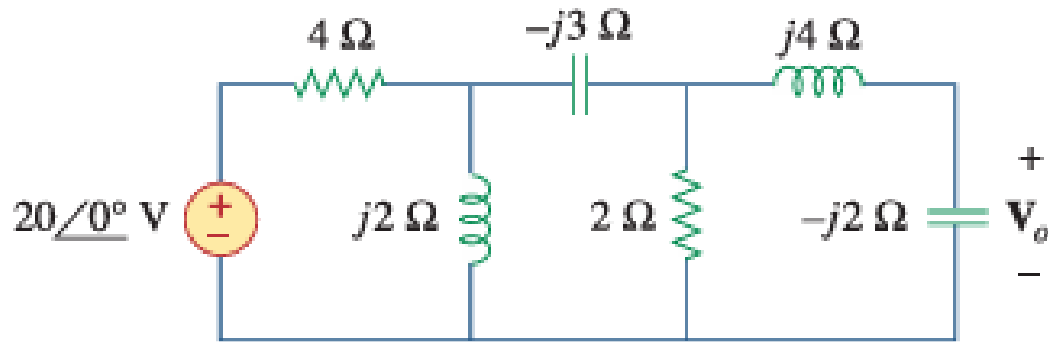
86. Find i_o in the circuit using Superposition theorem.



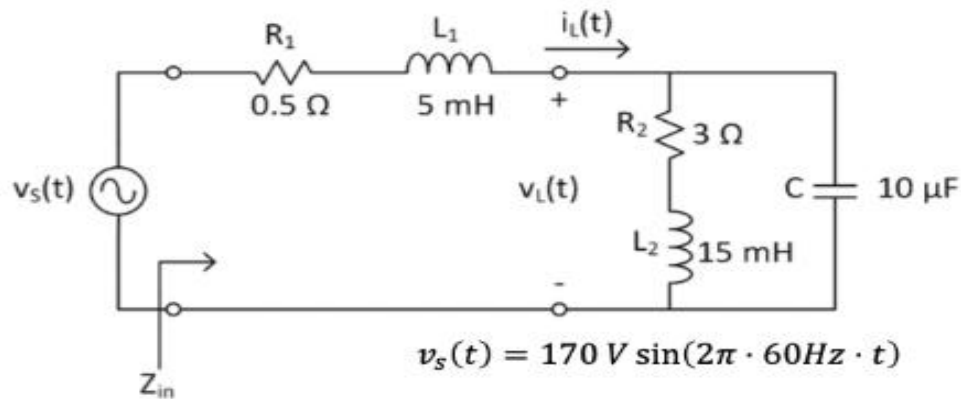
87. Find I_x in the circuit using the concept of Source transformation.



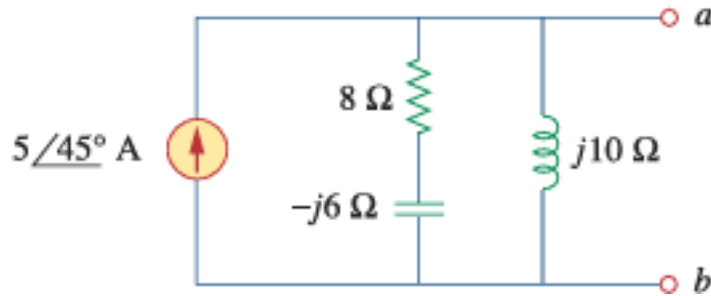
88. Find V_0 in the circuit using Source transformation.



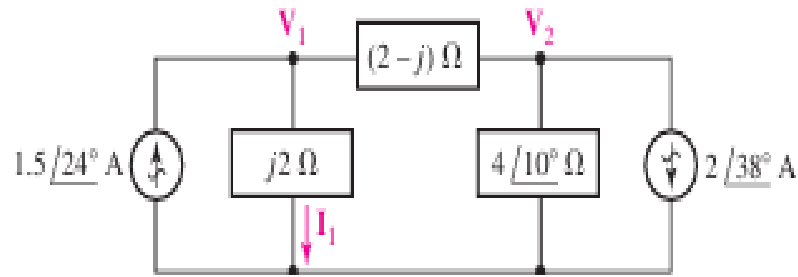
89. Consider the circuit shown, modelling a source driving a load through a cable. Using phasor analysis method determine a) the impedance at 60 Hz, b) voltage across the load $v_L(t)$, and c) current delivered to the load $i_L(t)$.



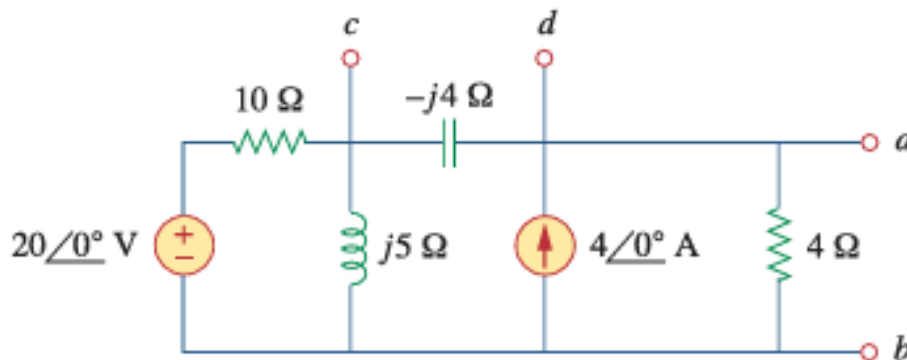
90. For the circuit in the figure, find the Thevenin equivalent at the terminals a-b.



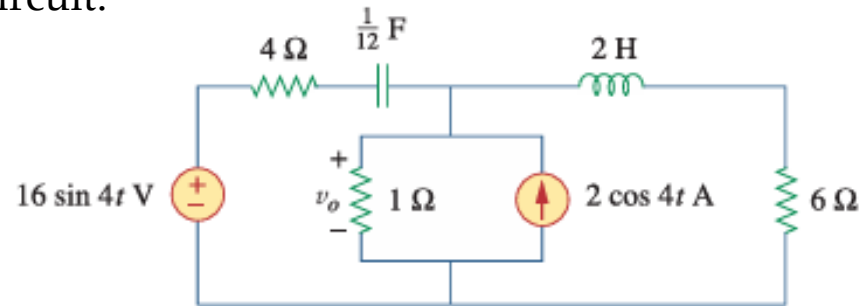
91. Obtain the Thevenin equivalent seen by the $(2 - j)\ \Omega$ impedance in the circuit and employ it to determine the current I_1 . Use phasor analysis method.



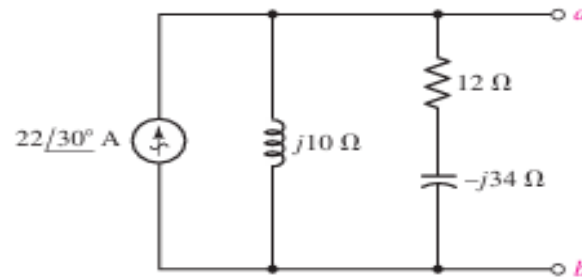
92. Find Thevenin equivalent of the circuit as seen from : (a) terminals a-b ,(b) terminals c-d.



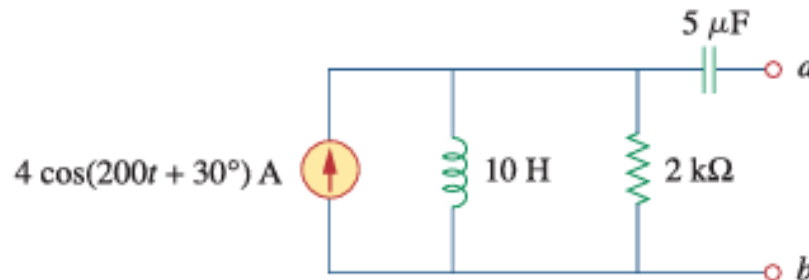
93. Determine v_o in the circuit.



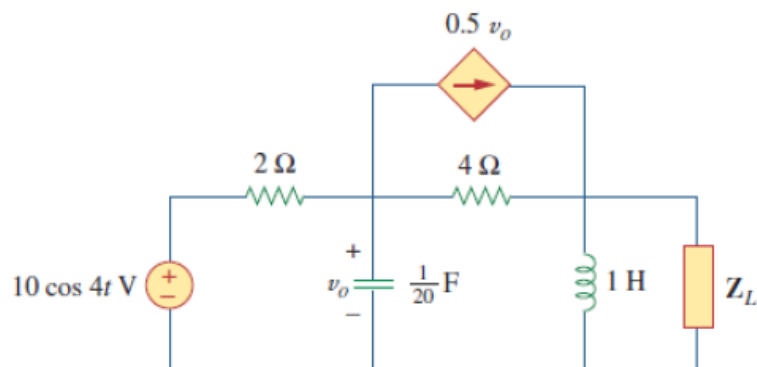
94. With regard to the circuit depicted in Fig. 7, a) determine the Norton equivalent seen looking into the terminals a-b, b) compute the current flowing from a to b if a $(7 - j2)$ impedance is connected across them. Use phasor analysis method.



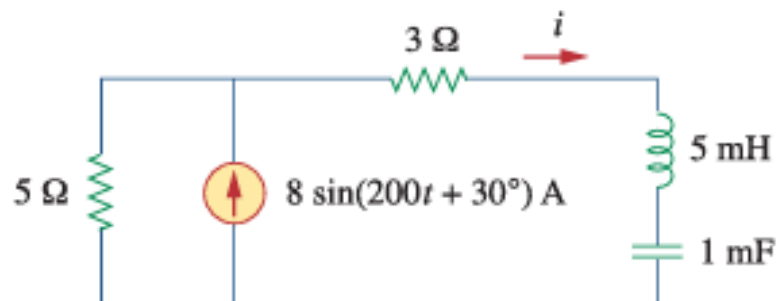
95. Determine the Norton equivalent of the circuit at terminals a-b.



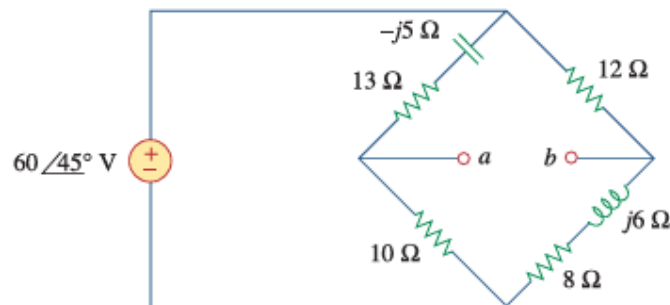
96. Determine the maximum average power delivered to the load for the circuit.



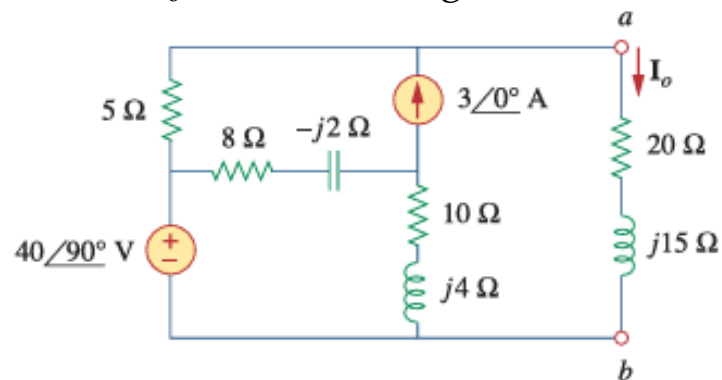
97. Using source transformation find i in the circuit given below.



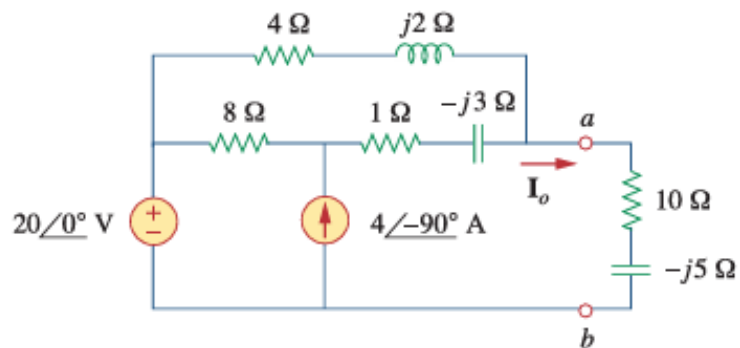
98. Determine the Norton equivalent of the circuit at terminals a-b.



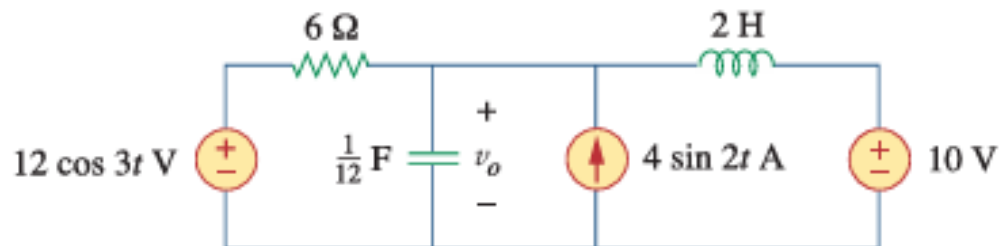
99. Find current I_0 in the circuit given below using Norton's theorem.



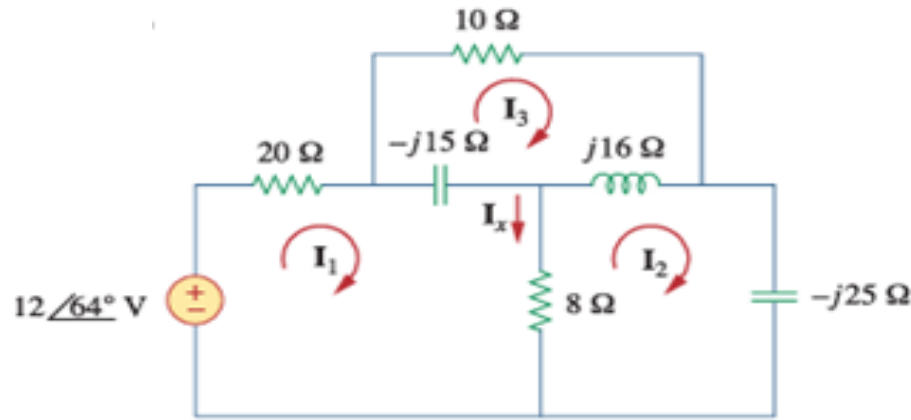
100. Determine the Norton equivalent of the circuit as seen from the terminals a-b. Use the equivalent circuit to find I_0 .



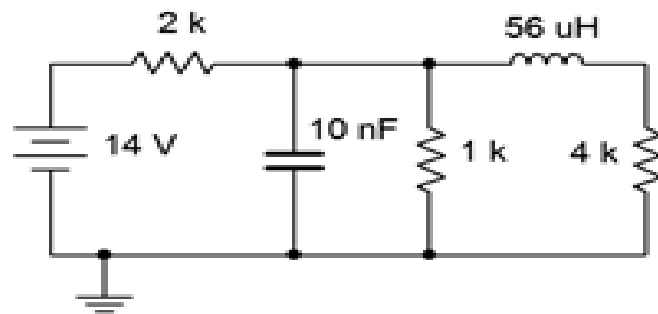
101. Solve for $v_o(t)$ in the circuit using superposition theorem.



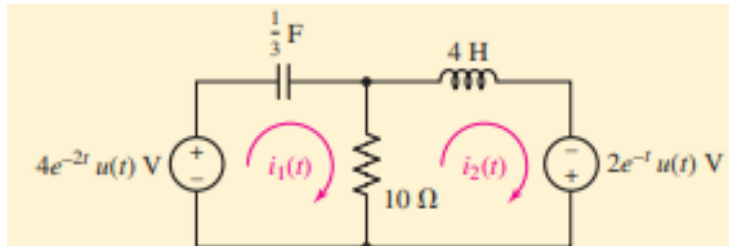
102. Find current I_1 , I_2 , I_3 , and I_x in the circuit given.



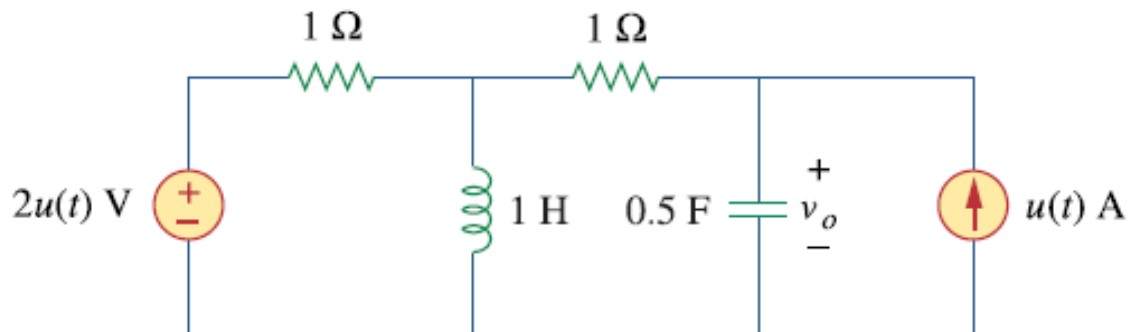
103. Assuming the initial current through the inductor is zero and the capacitor is uncharged in the circuit, determine the current through the $2\text{k}\Omega$ resistor when power is applied and after the circuit has reached steady state. Draw each of the equivalent circuits in the phasor domain.



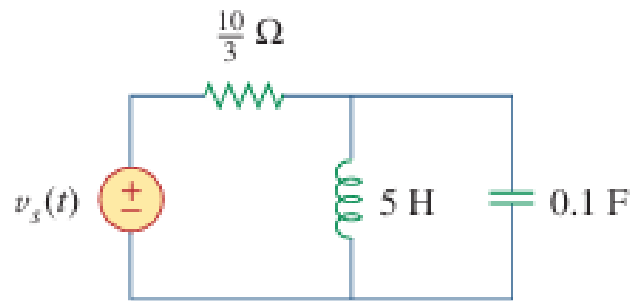
104. Determine the two mesh currents i_1 and i_2 in the circuit below. There is no energy initially stored in the circuit.



105. Find $v_o(t)$ for all time greater than zero in the circuit given below.



106. Consider the circuit below. Find the value of the voltage across the capacitor assuming that the value of $v_s(t) = 10 u(t)$ V and assume that at $t = 0$, -1 A flows through the inductor and +5 V is across the capacitor.



Module:4

Circuit Analysis in the S-domain: Circuit Element Models; Circuit Analysis with and without initial conditions; Transfer Functions; From Laplace to Fourier Transform- Frequency Domain Analysis of Linear Circuits; Passive Filters- Lowpass Filter, High Pass Filter, Bandpass Filter, Band Stop Filter.

107. Find the Laplace transform of each of the following functions:

(a) $t^2 \cos(2t + 30^\circ)u(t)$ (b) $3t^4 e^{-2t}u(t)$

(c) $2tu(t) - 4\frac{d}{dt}\delta(t)$ (d) $2e^{-(t-1)}u(t)$

(e) $5u(t/2)$ (f) $6e^{-t/3}u(t)$

(g) $\frac{d^n}{dt^n}\delta(t)$

108. Find the Laplace transform of the following signals:

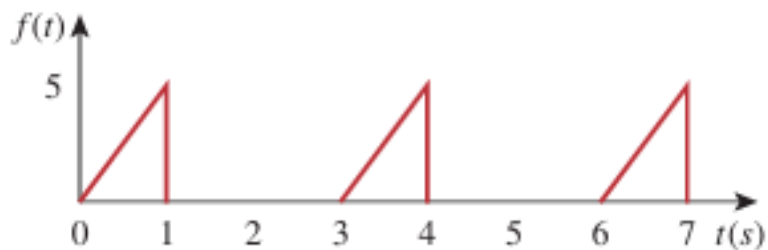
(a) $f(t) = (2t + 4)u(t)$

(b) $g(t) = (4 + 3e^{-2t})u(t)$

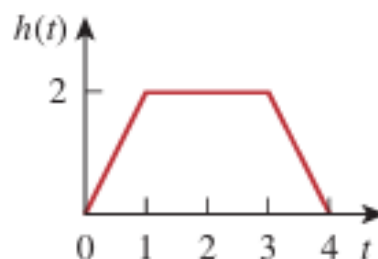
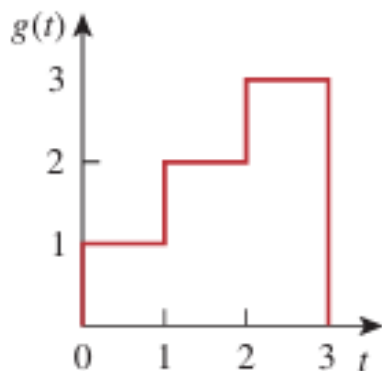
(c) $h(t) = (6 \sin(3t) + 8 \cos(3t))u(t)$

(d) $x(t) = (e^{-2t} \cosh(4t))u(t)$

109. Determine the Laplace transform of the function in Figure shown below.



110. Obtain the Laplace transforms of the functions in the figures below.



111. Determine the inverse Laplace transform of each of the following functions:

$$(a) \frac{8(s+1)(s+3)}{s(s+2)(s+4)}$$

$$(b) \frac{s^2 - 2s + 4}{(s+1)(s+2)^2}$$

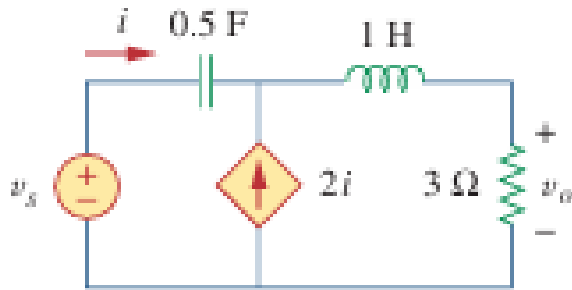
$$(c) \frac{s^2 + 1}{(s+3)(s^2 + 4s + 5)}$$

112. The transfer function of a system is $H(s) = (s+2) / (s^2 + 3s + 4)$. If the input is $4te^{-2t} u(t)$, find its output.

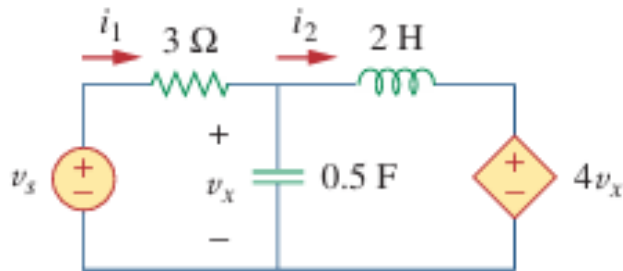
113. The transfer function of a certain circuit is given. Find the impulse response of the circuit.

$$H(s) = \frac{5}{s+1} - \frac{3}{s+2} + \frac{6}{s+4}$$

114. Obtain the transfer function $H(s) = V_o / V_s$ for the circuit given below.



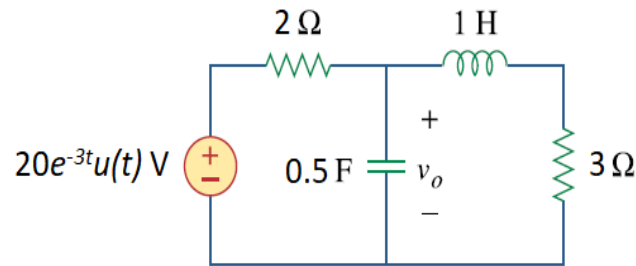
115. For the circuit given below, find: a) I_1 / V_s b) I_2 / V_x



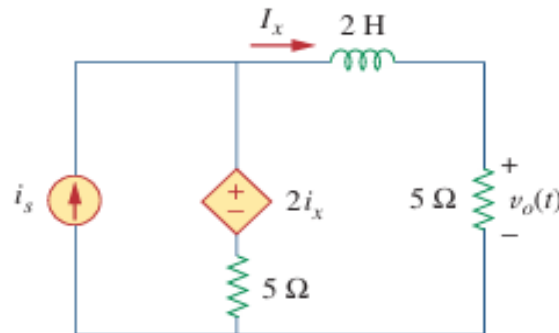
116. The transfer function of a system is given $H(s) = \frac{s^2}{3s + 1}$

Find the output when the system has an input of $4 e^{-t/3} u(t)$.

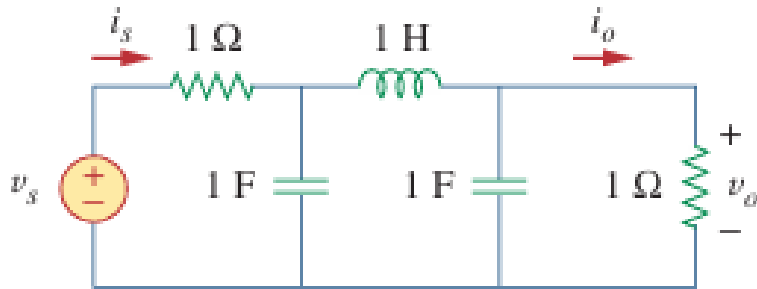
117. Apply the Thevenin theorem to determine $v_o(t)$ for $t > 0$ for the circuit given.



118. Assume that there is no initial energy stored in the circuit at $t = 0$ and that $i_s(t) = 10 u(t) \text{ A}$, a) find $V_o(s)$ using Thevenin's theorem. b) apply the initial and final value theorems to find $v_o(0^-)$ and $v_o(\infty)$ and c) determine $v_o(t)$.



119. Refer to the network given below. Find the following transfer functions:



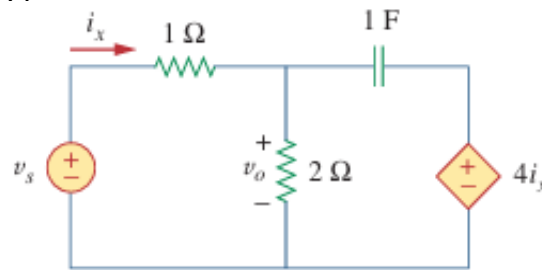
(a) $H_1(s) = V_o(s)/V_s(s)$

(b) $H_2(s) = V_o(s)/I_s(s)$

(c) $H_3(s) = I_o(s)/I_s(s)$

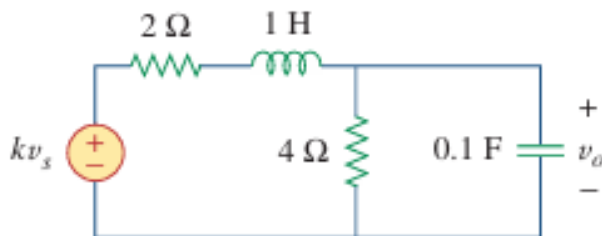
(d) $H_4(s) = I_o(s)/V_s(s)$

120. The initial energy in the circuit of Fig. 10 is zero at $t = 0$. Assume that $v_s = 30 u(t)$, a) find $V_o(s)$ using the Thevenin's theorem. b) apply the initial and final value theorems to find $v_o(0^-)$ and $v_o(\infty)$ and c) determine v_o''

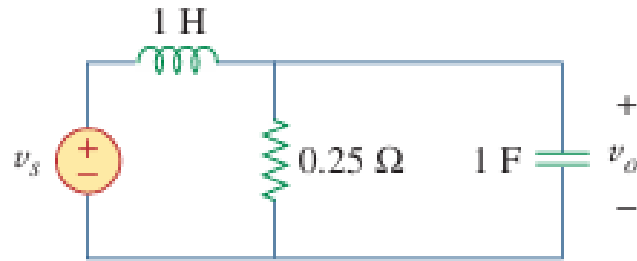


121. When the input to a system is a unit step function, the response is $10 \cos 2t u(t)$. Obtain the transfer function of the system.

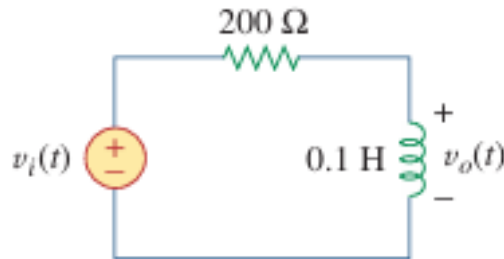
122. For the circuit below, find $H(s) = V_o(s)/V_s(s)$. Assume zero initial conditions.



123. Find the transfer function of the below. Show that the circuit is a lowpass filter.



124. Determine the type of filter and calculate the corner frequency f_c .

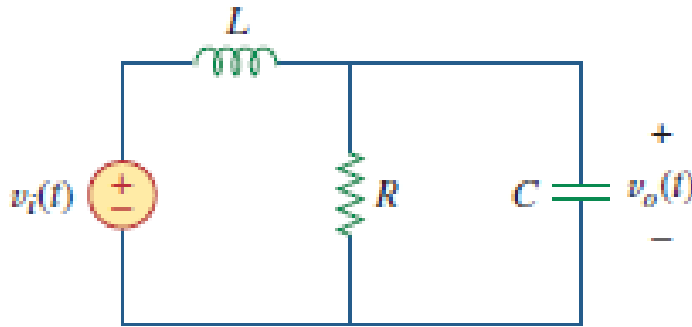


125. Determine the range of frequencies that will be passed by a series RLC bandpass filter with $R = 10\ \Omega$, $L = 25\text{ mH}$ and $C = 0.4\ \mu\text{F}$. Find the quality factor.

126. The tuner for an FM radio requires a band-pass filter with a centre frequency of 100 MHz (frequency of the FM station) and a bandwidth of 2 MHz . Design a filter circuit to accomplish this and evaluate its cut-off frequencies.

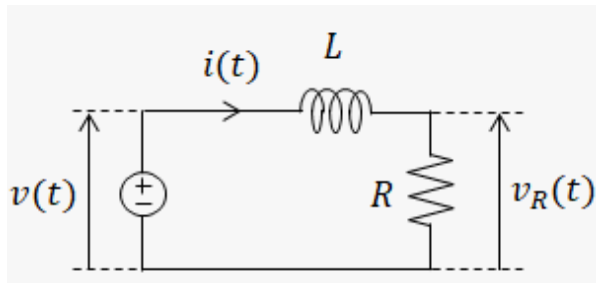
127. Design a bandpass filter with a lower cutoff frequency of 20.1 kHz and an upper cutoff frequency of 20.3 kHz. Take $R = 20 \text{ k}\Omega$. Calculate L , C , and Q .

128. Determine what type of filter is shown in Figure. Calculate the corner or cut-off frequency. Take $R = 2 \text{ k}\Omega$, $L = 2 \text{ H}$, and $C = 2 \text{ }\mu\text{F}$.

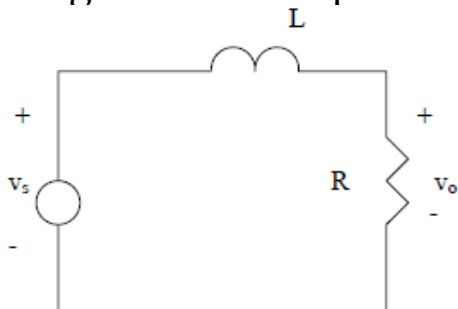


129. Design a filter to reject a 200 Hz sinusoid while passing other frequencies. Draw the circuit with the values of inductor (L) and capacitor (C) included in the circuit. Consider $R = 150 \text{ }\Omega$ and the bandwidth to be 100 Hz. Determine the quality factor of the circuit.

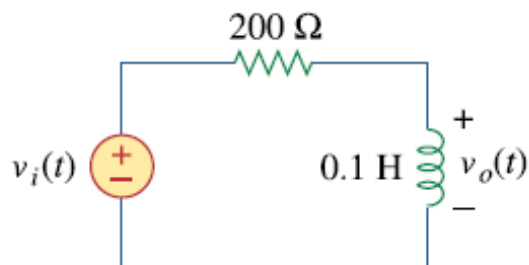
130. Show that a series LR circuit is a low pass filter if the output is taken across the resistor. Calculate the corner frequency if $L = 2 \text{ mH}$ and $R = 10 \text{ k}\Omega$



131. Design an RL low pass filter that uses a 40 mH coil and has a cut-off frequency of 5 kHz.

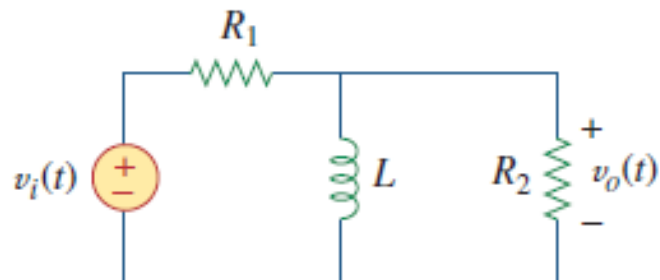


132. Determine what type of filter is in Figure. Calculate the corner frequency in Hz.



133. In a highpass RL filter with a cut-off frequency of 100 kHz, $L = 40\text{ mH}$, find R .

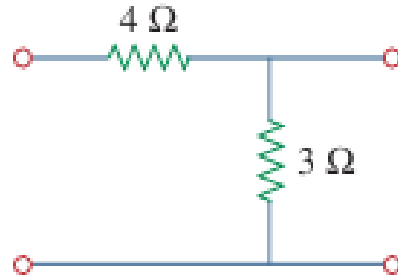
134. Identify the type of filter in figure. Obtain the transfer function and calculate the corner frequency if $R_1 = R_2 = 100\ \Omega$ and $L = 2\text{ mH}$.



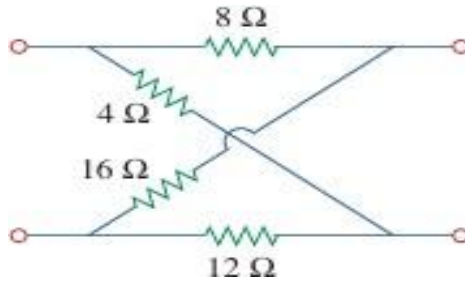
Module:5

Two-Port Networks and Network Graphs 10 hours Two-Port Networks-Impedance Parameters, Admittance Parameters, Hybrid Parameters, Transmission Parameters, Relationships Between Parameters, and Interconnection of Networks. Network Graphs-Incidence, Tie-Set, and Cut-Set Matrices for Circuit Analysis.

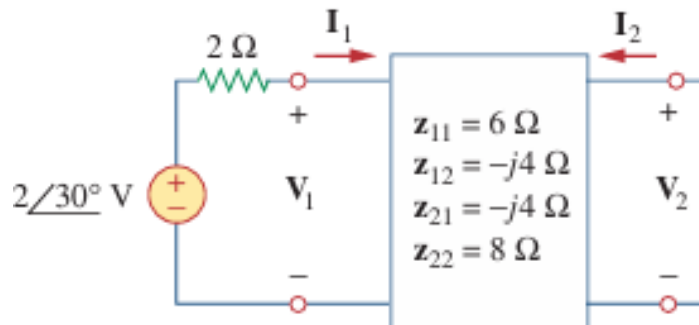
135. Find the z parameters of the two-port network in the figure below.



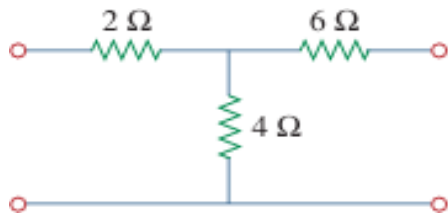
136. Determine the z parameters for the circuit.



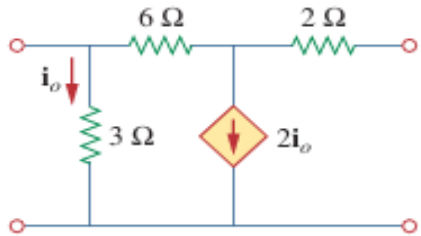
137. Calculate and in the two-port of the circuit shown below.



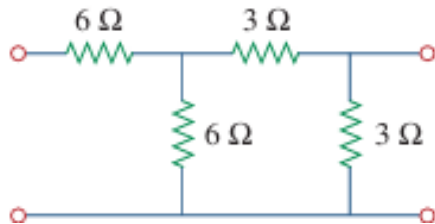
138. Obtain the y parameters for the T network shown in the figure below.



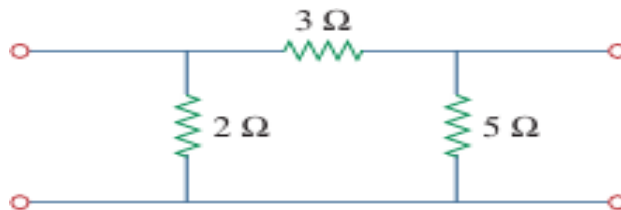
139. Obtain the y parameters for the circuit.



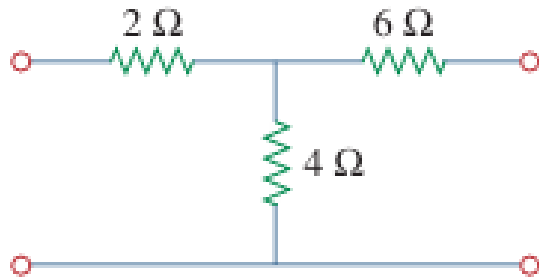
140. Calculate the y parameters for the two-port network given



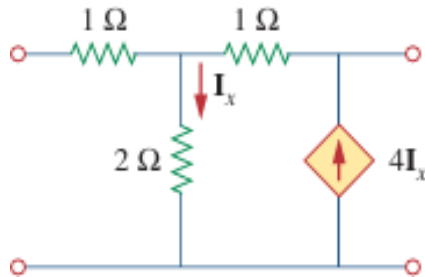
141. Determine the h parameters for the circuit.



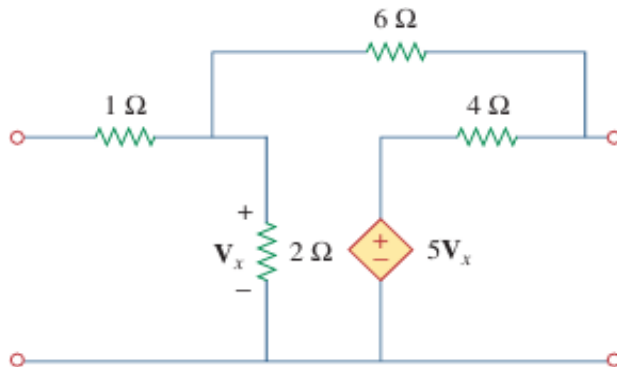
142. Find the transmission parameters for the circuit.



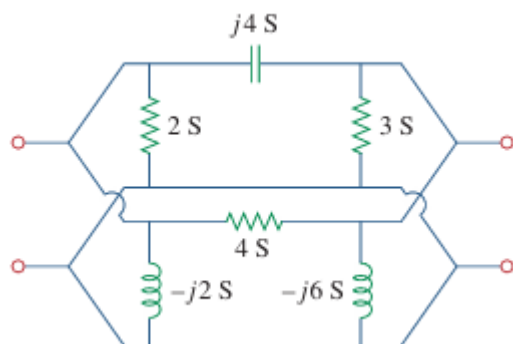
143. Find the transmission parameters for the circuit



144. Obtain the transmission parameters for the network



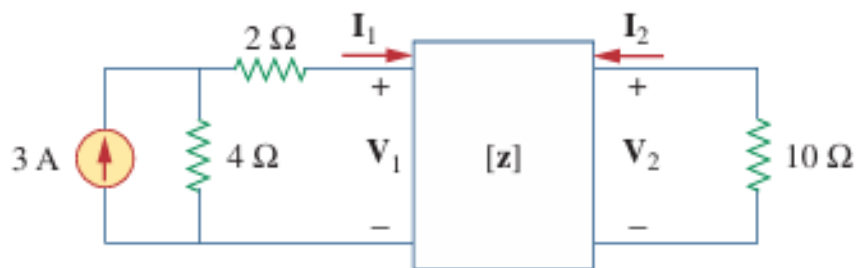
145. Find the y parameters of the two-port network shown below.



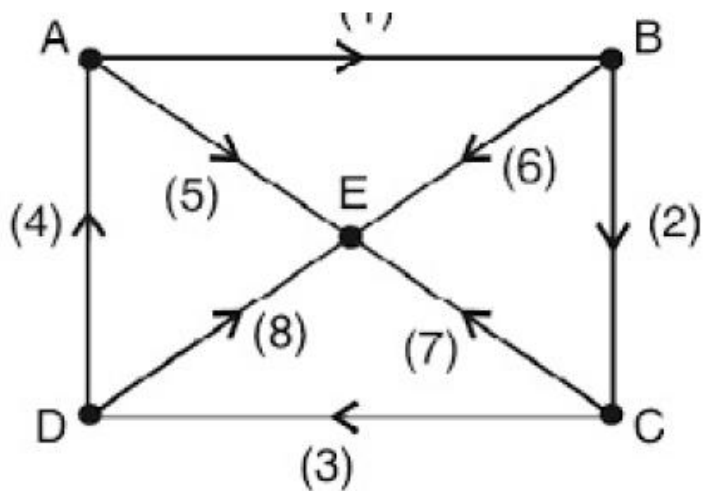
146. Determine a two-port network that is represented by the following zparameters:

$$[\mathbf{z}] = \begin{bmatrix} 6 + j3 & 5 - j2 \\ 5 - j2 & 8 - j \end{bmatrix} \Omega$$

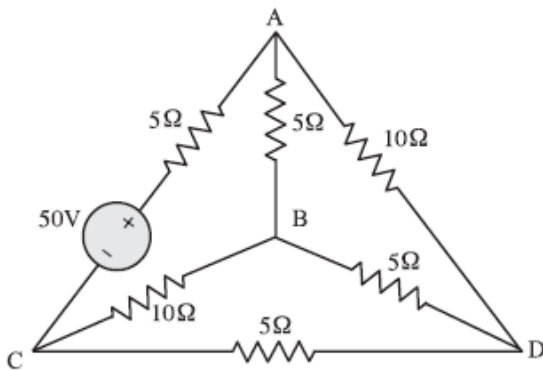
147. For the circuit shown: $[\mathbf{z}] = \begin{bmatrix} 10 & -6 \\ -4 & 12 \end{bmatrix} \Omega$ Find I_1 , I_2 , V_1 and V_2 .



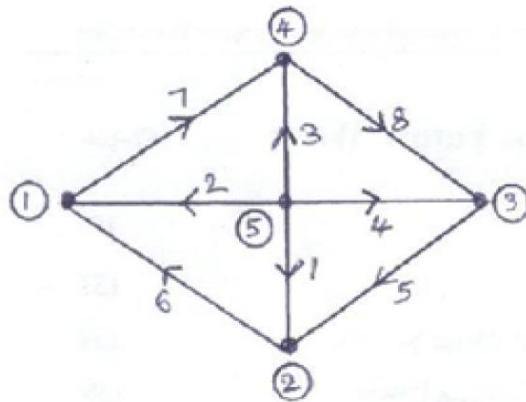
148. Form the tie-set matrix for the graph shown with respect to a tree.



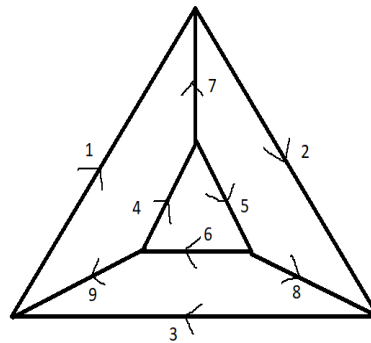
149. Form the tie-set matrix for the network given below with respect to a tree.



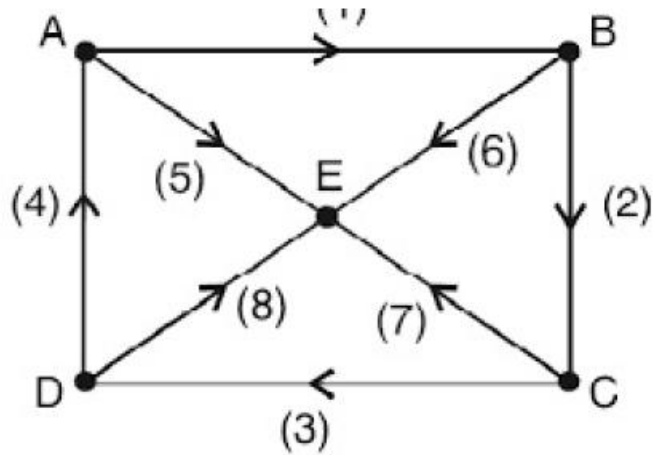
150. Construct the cut-set matrix for the tree comprising of branches 1, 2, 3 and 4 for the network shown in figure. Write down the corresponding voltage and current equations.



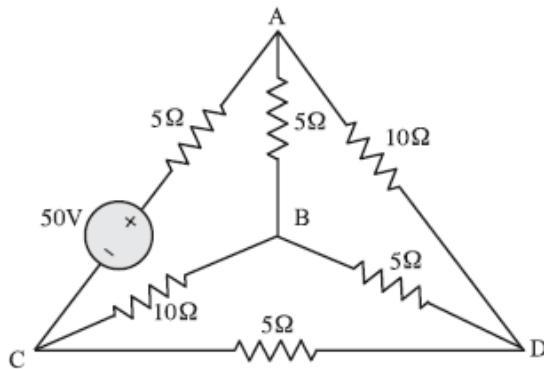
151. Construct the tie-set matrix for the tree comprising of branches a) 9, 4, 7, 5 and 8 b) 9, 1, 2, 8, and 5) in the network shown below. Write down the corresponding voltage and current equations.



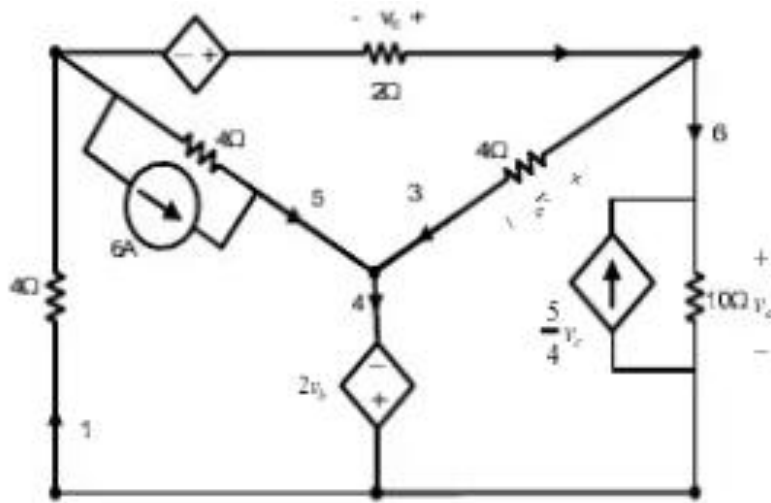
152. Find the Tie-set of the given graph.



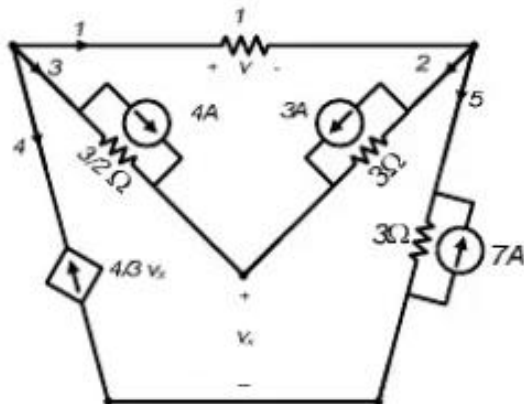
153. Form the tie-set matrix for the network given below with respect to a tree.



154. Write the KCL equations for the **fundamental loops** in the network.



155. Consider the branches of the network and draw the graph. Write the KVL equations for the tree with twigs 3, 4, 5.



156. In the following graph: a) choose a tree such that $(1, 3, 4, 6)$ and $(4, 5, 6, 8)$ are two of its fundamental cuts and $(1, 6, 2, 5, 7)$ is one of its fundamental loops. Then determine all other fundamental loops and fundamental cut-sets of the tree.

b) Determine a tree that has branches 1, 4, and 6 as its links (find all possible answer). C) determine a tree that has branches 1, 4, and 6 as its tree branches (find all possible answer).

